MATH 550
Topology
Lecture Notes

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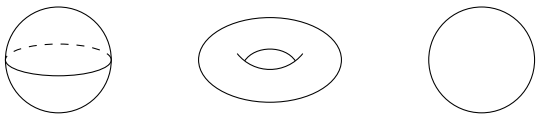
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Chapter 1

Sets, Relations, and Countability

1.1 Spaces and equivalence relations

A space is a set together with extra structure. A sphere, a torus, and a circle are examples of spaces; topology describes the structure that determines which subsets are open.



sphere torus circle

Definition 1.1. A *relation* on a set A is a subset $R \subseteq A \times A$. We write xRy when $(x, y) \in R$. An *equivalence relation* is a relation satisfying:

(i) xRx for every $x \in A$ (reflexivity);

(ii) xRy implies yRx (symmetry);

(iii) xRy and yRz imply xRz (transitivity).

For example, xRy if $x \geq y$ is a relation on $\mathbb{R}$, but it is not an equivalence relation: it is not symmetric.

Definition 1.2. Let $\sim$ be an equivalence relation on A. The *equivalence class* of $a \in A$ is

$$[a] = \{x \in A : x \sim a\}.$$

The *quotient set* is $A/\!\sim = \{[a] : a \in A\}$. The natural map $q : A \to A/\!\sim$ sends a to $[a]$. A *partition* of A is a collection of nonempty, pairwise disjoint subsets whose union is A.

Exercise 1.1. Show that the equivalence classes of an equivalence relation partition A, and that every partition determines an equivalence relation.

Solution. Reflexivity gives $a \in [a]$, so the classes are nonempty and cover A. Suppose $c \in [a] \cap [b]$. Then $c \sim a$ and $c \sim b$, hence $a \sim b$. If $x \in [a]$, transitivity gives $x \sim b$, so $[a] \subseteq [b]$. The reverse inclusion follows in the same way. Thus classes are equal or disjoint. Conversely, given a partition

$\mathcal{P}$, define $x \sim y$ when x, y belong to the same member of $\mathcal{P}$. Every point belongs to exactly one member. This proves reflexivity and symmetry; if x, y lie in one member and y, z in another, the common point y forces the members to coincide, proving transitivity. Its classes are exactly the members of $\mathcal{P}$.

Example 1.3. On $\mathbb{R}^2$, define

$$(x_1, y_1) \sim (x_2, y_2) \quad \Longleftrightarrow \quad x_1^2 + y_1^2 = x_2^2 + y_2^2.$$

This is an equivalence relation because equality of the two squared distances is reflexive, symmetric, and transitive. The points $(1, 1)$ and $(2, 2)$ are not related; $(1, 1)$ and $(-1, 1)$ are related. The corresponding partition is

$$\{C_r : r \geq 0\}, \qquad C_r = \{(x, y) : x^2 + y^2 = r^2\}.$$

For $r > 0$, C_r is a circle; $C_0 = \{(0, 0)\}$.

Example 1.4. On $\mathbb{Z}$, let $x \sim y$ if $x \equiv y \pmod 5$, or equivalently $5 \mid (x - y)$. Then

$$[0] = 5\mathbb{Z}, \qquad [1] = 1 + 5\mathbb{Z}, \qquad [6] = [1].$$

There are exactly five classes:

$$\mathbb{Z}/\!\sim \; = \{[0], [1], [2], [3], [4]\}.$$

The map $q(n) = [n]$ is surjective and identifies all integers with the same remainder. In particular, $q(1) = [1]$, $q(2) = [2]$, and $q(-3) = [2]$, since $-3 - 2 = -5$ is divisible by 5.

Definition 1.5. A *strict total order* $<$ on A is a transitive relation such that, for distinct x, y, exactly one of $x < y$ and $y < x$ holds, and $x < x$ never holds. We also write $x \leq y$ for $x < y$ or $x = y$.

For example, the lexicographic order on ordered pairs is defined by

$$(a, b) < (c, d) \quad \Longleftrightarrow \quad a < c \text{ or } (a = c \text{ and } b < d).$$

1.2 Finite and countable sets

Write $\mathbb{Z}_+ = \{1, 2, 3, \ldots\}$ and $\mathbb{N} = \{0, 1, 2, \ldots\}$. Recall that a function is *injective* if equal outputs force equal inputs, *surjective* if every element of its codomain is attained, and *bijective* if both hold.

Definition 1.6. A set A is *finite* if $A = \varnothing$ or there is a bijection $\{1, \ldots, n\} \to A$ for some $n \in \mathbb{Z}_+$. It is *infinite* otherwise. It is *countably infinite* if there is a bijection $\mathbb{Z}_+ \to A$, and *countable* if it is finite or countably infinite.

Exercise 1.2. Establish the following facts about finite sets.

(i) An injection from A into a finite set implies that A is finite.

(ii) A surjection from a finite set onto A implies that A is finite.

(iii) If $\{1, \dots, m\}$ and $\{1, \dots, n\}$ are in bijection, then $m = n$.

(iv) If $B \subseteq A$ and A is finite, then $|B| \leq |A|$; if the inclusion is proper, then $|B| < |A|$.

(v) Finite unions and finite Cartesian products of finite sets are finite.

Solution. A subset of $\{1, \dots, n\}$ can be listed in increasing order, giving a bijection with $\{1, \dots, k\}$ for some $0 \leq k \leq n$; here $k = 0$ denotes the empty set. An injection identifies its domain with its image, proving (i). For a surjection $f : \{1, \dots, n\} \to A$, choose for each $a \in A$ the least j with $f(j) = a$. These chosen indices give an injection $A \to \{1, \dots, n\}$, proving (ii).

For (iii), the pigeonhole principle follows by induction: an injection from an $(n+1)$-element set into an n-element set cannot exist. Indeed, delete one input and its output; after relabeling, this would give an injection from an n-element set into an $(n-1)$-element set. A bijection between the two displayed sets therefore gives $m \leq n$ and $n \leq m$.

For (iv), transport B by a bijection from A to $\{1, \dots, n\}$ and use its increasing enumeration. If a point is omitted, at most $n - 1$ indices remain. For (v), list the elements of each set and remove repetitions to list a finite union. For two factors of sizes m, n, list the pairs in rows; there are mn pairs. Induction gives both assertions for any finite number of sets.

Note 1.7 (Well-ordering principle). Every nonempty subset of $\mathbb{Z}_+$ has a smallest element. This permits recursive increasing enumeration of infinite subsets of $\mathbb{Z}_+$.

Lemma 1.8. *Every infinite subset $S \subseteq \mathbb{Z}_+$ is countably infinite.*

Proof. Let $s_1 = \min S$, and recursively define

$$s_{n+1} = \min(S \setminus \{s_1, \dots, s_n\}).$$

The remaining set is nonempty because S is infinite. The sequence is strictly increasing, so the map $n \mapsto s_n$ is injective. If some $s \in S$ never appeared, every s_n would be less than s; this would put infinitely many distinct positive integers in the finite set $\{1, \dots, s-1\}$. Thus every s appears, proving surjectivity.

For example, the primes can be listed as $2, 3, 5, 7, \dots$, each time selecting the least unused prime.

Theorem 1.9. *For a nonempty set A, the following are equivalent:*

(i) A is countable;

(ii) there is a surjection $\mathbb{Z}_+ \to A$;

(iii) there is an injection $A \to \mathbb{Z}_+$.

Exercise 1.3. Prove Theorem 1.9.

Solution. If A is countably infinite, a bijection proves both (ii) and (iii). If A has $k \geq 1$ elements, list them as $a_1, \ldots, a_k$. The map taking $j \leq k$ to a_j and all $j > k$ to a_1 is surjective; the map $a_j \mapsto j$ is injective.

Given a surjection $f : \mathbb{Z}_+ \to A$, define $g(a) = \min f^{-1}(\{a\})$. Distinct points have disjoint fibers, so g is injective. Given an injection $g : A \to \mathbb{Z}_+$, it is a bijection onto $g(A)$. If $g(A)$ is finite, so is A; if it is infinite, Lemma 1.8 gives a bijection $\mathbb{Z}_+ \to g(A)$, and composition with g^{-1} gives a bijection $\mathbb{Z}_+ \to A$.

Example 1.10 (Diagonal enumeration)**.** The set $\mathbb{Z}_+ \times \mathbb{Z}_+$ is countably infinite. List pairs by successive diagonals $i + j = 2, 3, 4, \ldots$:

$$(1,1),\ (1,2),(2,1),\ (1,3),(2,2),(3,1),\ldots.$$

Every pair appears once. More explicitly, the index of (i, j) in this listing is

$$\frac{(i+j-2)(i+j-1)}{2} + i.$$

Thus the diagonal enumeration is a bijection with $\mathbb{Z}_+$. The map $n \mapsto n - 1$ also shows that $\mathbb{N}$ is countably infinite.

Example 1.11. The map $(a, b) \mapsto a/b$ from $\mathbb{Z}_+ \times \mathbb{Z}_+$ onto $\mathbb{Q}_{>0}$ is surjective. Hence $\mathbb{Q}_{>0}$ is countable, and is infinite because it contains $\mathbb{Z}_+$. Listing positive rationals and their negatives, together with 0, shows that $\mathbb{Q}$ is countably infinite.

Theorem 1.12. *A countable union of countable sets is countable. A finite Cartesian product of countable sets is countable.*

Proof. For the union, discard empty sets. If every set is empty, the assertion is immediate. For the nonempty sets, choose enumerations $f_i : \mathbb{Z}_+ \to A_i$ that are surjective; finite sets may be enumerated with repetitions. As usual, this choice of a countable family of enumerations is made in the ordinary set-theoretic setting with the axiom of choice. If there are infinitely many nonempty sets, then

$$g : \mathbb{Z}_+ \times \mathbb{Z}_+ \longrightarrow \bigcup_{i \geq 1} A_i, \qquad g(i,j) = f_i(j)$$

is surjective. Compose with the diagonal enumeration. If there are only finitely many sets, repeat one of them to make an infinite indexed family; the same argument applies.

For products, if a factor is empty, the product is empty. Otherwise choose surjections $f : \mathbb{Z}_+ \to A$, $g : \mathbb{Z}_+ \to B$. Then $(i, j) \mapsto (f(i), g(j))$ is onto $A \times B$, which is therefore countable. Induction proves the assertion for every finite number of factors.

Theorem 1.13 (Cantor)**.** $\mathbb{R}$ *is uncountable.*

Exercise 1.4. Prove Cantor's theorem by a diagonal argument. Show also why a countably infinite product of countable sets need not be countable.

Solution. If $(0, 1)$ were countable, list its elements as $x_1, x_2, \ldots$ and write

$$x_n = 0.a_{n1}a_{n2}a_{n3}\ldots$$

using the decimal expansion that is not eventually 9. Choose $b_n = 1$ if $a_{nn} \neq 1$, and $b_n = 2$ if $a_{nn} = 1$. Then $y = 0.b_1 b_2 \ldots$ belongs to $(0, 1)$ and differs from x_n in the nth decimal digit. Its digits are all 1 or 2, so decimal ambiguity cannot identify it with a listed number. This contradicts the listing. Thus $(0, 1)$, and hence $\mathbb{R}$, is uncountable.

If all binary sequences could be listed as $s^{(1)}, s^{(2)}, \ldots$, define $t_n = 1 - s_n^{(n)}$. The sequence t differs from $s^{(n)}$ at coordinate n and is missing from the list. Consequently $\{0, 1\}^{\mathbb{Z}_+}$ is uncountable, even though each factor is finite. Thus X^n is countable for countable X and finite n, but $X^{\mathbb{Z}_+}$ need not be.

Chapter 2

Topologies and Bases

2.1 Topological spaces

Definition 2.1. A *topology* on a set X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ such that

(i) $\varnothing, X \in \mathcal{T}$;

(ii) the union of any subcollection of $\mathcal{T}$ belongs to $\mathcal{T}$;

(iii) the intersection of any finite subcollection of $\mathcal{T}$ belongs to $\mathcal{T}$.

The pair $(X, \mathcal{T})$ is a *topological space*. Members of $\mathcal{T}$ are *open sets*. A subset $F \subseteq X$ is *closed* if $X \setminus F$ is open.

It suffices in (iii) to require closure under the intersection of two open sets, together with $X \in \mathcal{T}$; induction gives all finite intersections. An arbitrary intersection of open sets need not be open.

Example 2.2. For $X = \{a, b, c\}$,

$$\mathcal{T} = \{\varnothing, \{a\}, \{a, b\}, X\}$$

is a topology. Its members are nested, so every nonempty union or finite intersection is again a member. In contrast,

$$\{\varnothing, \{a\}, \{b\}, \{c\}, X\}$$

is not a topology, since $\{a\} \cup \{b\} = \{a, b\}$ is absent.

Exercise 2.1. Let $X = \{a, b, c\}$. Is

$$\mathcal{U} = \{\varnothing, \{a, b\}, \{b, c\}, \{c\}\}$$

a topology? Does adjoining X suffice to make it a topology?

Solution. $\mathcal{U}$ is not a topology because $X \notin \mathcal{U}$. Adjoining X still does not suffice: the intersection $\{a, b\} \cap \{b, c\} = \{b\}$ is missing. In addition, $\{a, b\} \cap \{c\} = \varnothing$ is present, but closure must hold for every pair, so a single missing intersection disproves the assertion.

Example 2.3 (Discrete and indiscrete topologies)**.** On any set X, the *indiscrete* (or *antidiscrete*) topology is $\{\varnothing, X\}$, and the *discrete* topology is $\mathcal{P}(X)$. For $X = \{a, b\}$, there are exactly four topologies:

$$\{\varnothing, X\}, \qquad \{\varnothing, \{a\}, X\}, \qquad \{\varnothing, \{b\}, X\}, \qquad \mathcal{P}(X).$$

Indeed, a topology must contain $\varnothing$ and X; it may include either, both, or neither of the two remaining subsets. Each resulting collection satisfies the axioms. On $\{a, b, c\}$ the indiscrete and discrete topologies remain examples; the latter contains all eight subsets. If a topology contains all three singletons, closure under unions forces it to be discrete.

Definition 2.4. The *finite complement* or *cofinite topology* on X is

$$\mathcal{T}_{\text{cf}} = \{\varnothing\} \cup \{U \subseteq X : X \setminus U \text{ is finite}\}.$$

Proposition 2.5. *$\mathcal{T}_{\text{cf}}$ is a topology.*

Proof. $\varnothing$ belongs by definition, and X belongs because its complement is empty. For a family (U_α), if every member is empty the union is empty. Otherwise choose a nonempty U_{α_0}. Then

$$X \setminus \bigcup_\alpha U_\alpha \subseteq X \setminus U_{\alpha_0},$$

so the complement of the union is finite. For two nonempty members U, V, De Morgan's law gives

$$X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V),$$

a finite union of finite sets. If either member is empty, the intersection is empty. This proves all axioms.

Example 2.6. In discrete $\mathbb{R}$, the sets $[0, 1]$, $(0, 1)$, and $\mathbb{R} \setminus \{1\}$ are open and closed. In cofinite $\mathbb{R}$, $[0, 1]$ and $(0, 1)$ are neither open nor closed, because both they and their complements are infinite. The set $\mathbb{R} \setminus \{1\}$ is open but not closed; $\{5\}$ is closed but not open.

Exercise 2.2. Show that

$$\mathcal{T}_{\text{cc}} = \{\varnothing\} \cup \{U \subseteq X : X \setminus U \text{ is countable}\}$$

is a topology, called the *countable complement topology.*

Solution. Again $\varnothing, X$ belong. A union containing a nonempty member has complement contained in that member's countable complement, so the union is open. For nonempty U, V, the complement of $U \cap V$ is $(X \setminus U) \cup (X \setminus V)$, a countable set. Empty intersections with an empty member cause no difficulty. Thus the topology axioms hold. If X itself is countable, every subset has countable complement, and this topology is discrete.

2.2 A basis and the topology it generates

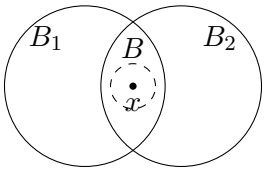
Definition 2.7. A collection $\mathcal{B}$ of subsets of X is a *basis for a topology* if

(i) $\bigcup_{B \in \mathcal{B}} B = X$;

(ii) whenever $x \in B_1 \cap B_2$ with $B_1, B_2 \in \mathcal{B}$, there is $B \in \mathcal{B}$ such that $x \in B \subseteq B_1 \cap B_2$.

The topology *generated by* $\mathcal{B}$ is

$$\mathcal{T}_{\mathcal{B}} = \{U \subseteq X : \text{for every } x \in U \text{ there is } B \in \mathcal{B} \text{ with } x \in B \subseteq U\}.$$



Theorem 2.8. *$\mathcal{T}_{\mathcal{B}}$ is a topology, and its elements are exactly the unions of members of $\mathcal{B}$.*

Proof. The condition for $\varnothing$ holds vacuously; the covering condition gives $X \in \mathcal{T}_{\mathcal{B}}$. If $x \in \bigcup_\alpha U_\alpha$, choose an index with $x \in U_\alpha$. There is $B \in \mathcal{B}$ with $x \in B \subseteq U_\alpha \subseteq \bigcup_\alpha U_\alpha$, so arbitrary unions are open. If $x \in U \cap V$, choose $B_1, B_2 \in \mathcal{B}$ with $x \in B_1 \subseteq U$ and $x \in B_2 \subseteq V$. The basis axiom gives B with

$$x \in B \subseteq B_1 \cap B_2 \subseteq U \cap V.$$

Thus $U \cap V$ is open.

Every basis element is open: for a point of B use B itself. Hence every union of basis elements is open. Conversely, for an open U choose such a B_x at each $x \in U$; then $U = \bigcup_{x \in U} B_x$. Equivalently, one may take the union of all basis elements contained in U.

Example 2.9. Open intervals form a basis on $\mathbb{R}$. Their intersection is either empty or another open interval, and they cover $\mathbb{R}$. The generated topology is the *usual* or *Euclidean topology*. On an arbitrary X, the singletons $\{\{x\} : x \in X\}$ form a basis. They cover X. If $x \in \{a\} \cap \{b\}$, then $x = a = b$, so the intersection is the basis element $\{x\}$ itself. This verifies the second basis axiom. Every subset of X is a union of singletons, so the generated topology is discrete. The collection $\mathcal{P}(X)$ is another basis generating the same topology. Thus distinct bases may generate the same topology.

Example 2.10. Open rectangles $(a, b) \times (c, d)$ form a basis on $\mathbb{R}^2$: they cover the plane, and nonempty intersections of two such rectangles are open rectangles. Open disks also form a basis.

Exercise 2.3. Verify the disk basis condition using the triangle inequality, and show that open disks and open rectangles generate the same topology on $\mathbb{R}^2$.

Solution. If $x \in B(p, r) \cap B(q, s)$, choose

$$0 < \varepsilon < \min\{r - \|x - p\|, s - \|x - q\|\}.$$

For $\|y - x\| < \varepsilon$, the triangle inequality gives $\|y - p\| < \varepsilon + \|x - p\| < r$ and similarly $\|y - q\| < s$. Hence $B(x, \varepsilon)$ lies in the intersection.

For a point $x = (x_1, x_2)$ in a rectangle $(a, b) \times (c, d)$, take a radius smaller than each of $x_1 - a, b - x_1, x_2 - c, d - x_2$. The disk with this radius lies in the rectangle. Conversely, if $x \in B(p, r)$, choose $\delta > 0$ with $\sqrt{2}\delta < r - \|x - p\|$. The rectangle

$$(x_1 - \delta, x_1 + \delta) \times (x_2 - \delta, x_2 + \delta)$$

lies in $B(p, r)$ because its points have distance less than $\sqrt{2}\delta$ from x. Each basic neighborhood for either collection thus contains a basic neighborhood for the other. The generated open sets coincide.

Lemma 2.11. *Let $(X, \mathcal{T})$ be a space and $\mathcal{B} \subseteq \mathcal{T}$. Then $\mathcal{B}$ is a basis for $\mathcal{T}$ if and only if, for each $U \in \mathcal{T}$ and $x \in U$, there is $B \in \mathcal{B}$ with $x \in B \subseteq U$.*

Proof. Necessity follows from Theorem 2.8. Conversely, apply the stated condition to $U = X$ to see that $\mathcal{B}$ covers X. If $x \in B_1 \cap B_2$, the intersection is $\mathcal{T}$-open, so the condition supplies a B inside it. Thus $\mathcal{B}$ satisfies the basis axioms. Every union of its members is $\mathcal{T}$-open, and every $\mathcal{T}$-open set is such a union by the condition. Hence the topology it generates is exactly $\mathcal{T}$.

2.3 Comparing topologies and the lower-limit topology

Definition 2.12. For topologies $\mathcal{T}, \mathcal{T}'$ on the same set X, $\mathcal{T}'$ is *finer* (or *larger*) than $\mathcal{T}$ if $\mathcal{T} \subseteq \mathcal{T}'$. Then $\mathcal{T}$ is *coarser* than $\mathcal{T}'$. If neither contains the other, they are *not comparable*.

The discrete topology is the finest topology on X, and the indiscrete topology is the coarsest.

Lemma 2.13 (Comparing bases)**.** *Suppose $\mathcal{B}$ and $\mathcal{B}'$ generate topologies $\mathcal{T}$ and $\mathcal{T}'$ on X. Then $\mathcal{T}'$ is finer than $\mathcal{T}$ if and only if, for every $B \in \mathcal{B}$ and every $x \in B$, there exists $B' \in \mathcal{B}'$ such that*

$$x \in B' \subseteq B.$$

Exercise 2.4. Prove the comparison criterion.

Solution. If $\mathcal{T} \subseteq \mathcal{T}'$, then each $B \in \mathcal{B}$ is $\mathcal{T}'$-open. The definition of a basis for $\mathcal{T}'$ gives the required B' at each $x \in B$. Conversely, the stated condition makes every $B \in \mathcal{B}$ open in $\mathcal{T}'$. Every $\mathcal{T}$-open set is a union of such B, hence is also $\mathcal{T}'$-open.

Definition 2.14. The *lower-limit topology* on $\mathbb{R}$ is generated by

$$\mathcal{B}_\ell = \{[a, b) : a, b \in \mathbb{R},\ a < b\}.$$

The resulting space is the *Sorgenfrey line*, denoted $\mathbb{R}_\ell$.

Exercise 2.5. Verify that $\mathcal{B}_\ell$ is a basis and that the lower-limit topology is strictly finer than the usual topology on $\mathbb{R}$.

Solution. Each x lies in $[x, x+1)$, so the basis covers $\mathbb{R}$. The nonempty intersection of $[a, b)$ and $[c, d)$ is

$$[\max\{a, c\}, \min\{b, d\}),$$

again a basis element. If $x \in (a, b)$, then $x \in [x, b) \subseteq (a, b)$, so every usual open interval is lower-limit open. Since usual open sets are unions of open intervals, every usual open set is lower-limit open. Finally, $[0, 1)$ is a lower-limit basis element but is not usually open: no usual open interval around 0 lies in $[0, 1)$. The inclusion of topologies is strict.

Chapter 3

Order, Product, and Subspace Topologies

3.1 The order topology

Suppose $(X, <)$ is a totally ordered set. Define

$$(a, b) = \{x : a < x < b\}, \quad [a, b) = \{x : a \leq x < b\}, \quad (a, b] = \{x : a < x \leq b\},$$

and define $[a, b]$ similarly.

Definition 3.1. For a totally ordered set with at least two points, the *order topology* is generated by all intervals (a, b), together with $[m, b)$ when X has a least element m, and $(a, M]$ when X has a greatest element M. A one-point ordered set has the topology $\{\varnothing, X\}$.

Exercise 3.1. Show that these intervals form a basis. Describe it for $\mathbb{R}$, $[0, 1)$, and $(0, 1]$ with their usual orders.

Solution. A point that is neither least nor greatest has an element below it and an element above it, so it lies in an interval (a, b). A least point lies in $[m, b)$ for any $b > m$, and a greatest point lies in $(a, M]$ for any $a < M$. Hence the collection covers X. A nonempty intersection of two basic intervals has the stronger of their lower bounds and the stronger of their upper bounds, retaining a closed endpoint only if it is the least or greatest point of X. It is therefore another interval of the listed type. This proves the intersection axiom.

On $\mathbb{R}$, there are no endpoints, so the open intervals generate the usual topology. On $[0, 1)$, use intervals (a, b) with $0 \leq a < b < 1$ and initial intervals $[0, b)$ with $0 < b < 1$. These cover every point, including those arbitrarily close to 1. On $(0, 1]$, use interior intervals and terminal intervals $(a, 1]$ with $0 < a < 1$. The one-sided intervals supply neighborhoods of the included endpoint.

3.2 Subbases and products

Definition 3.2. A *subbasis* $\mathcal{S}$ for a topology on X is a collection of subsets whose finite intersections form a basis. We include the empty intersection, interpreted as X. The generated open sets are arbitrary unions of these finite intersections.

Example 3.3. On $\mathbb{R}$, the open rays $(-\infty, a)$ and (a, ∞) form a subbasis for the usual topology, since

$$(a, b) = (a, \infty) \cap (-\infty, b).$$

Finite intersections of rays are intervals, rays, $\mathbb{R}$, or $\varnothing$, all of which are usually open.

Definition 3.4. If X, Y are spaces, the *product topology* on $X \times Y$ has basis

$$\{U \times V : U \text{ open in } X,\ V \text{ open in } Y\}.$$

Exercise 3.2. Verify that this is a basis. If $\mathcal{B}, \mathcal{C}$ are bases of X, Y, show that

$$\mathcal{D} = \{B \times C : B \in \mathcal{B},\ C \in \mathcal{C}\}$$

is a basis for the product topology.

Solution. $X \times Y$ itself is an open rectangle, so the collection covers. Moreover,

$$(U_1 \times V_1) \cap (U_2 \times V_2) = (U_1 \cap U_2) \times (V_1 \cap V_2),$$

which is again an open rectangle. For the second assertion, let $(x, y) \in U \times V$. Choose $B \in \mathcal{B}$, $C \in \mathcal{C}$ with

$$x \in B \subseteq U, \qquad y \in C \subseteq V.$$

Then $(x, y) \in B \times C \subseteq U \times V$. Since $B \times C$ is product-open, Lemma 2.11 proves that $\mathcal{D}$ is a basis.

Proposition 3.5. *The collection*

$$\mathcal{S} = \{U \times Y : U \text{ open in } X\} \cup \{X \times V : V \text{ open in } Y\}$$

is a subbasis for the product topology.

Proof. Every member is product-open, and each basic rectangle is the finite intersection

$$U \times V = (U \times Y) \cap (X \times V).$$

Thus every product-open set is a union of finite intersections of members of $\mathcal{S}$, and these intersections are all product-open. The two generated topologies coincide.

3.3 Subspaces

Definition 3.6. For a space $(X, \mathcal{T})$ and a subset $Y \subseteq X$, the *subspace topology* on Y is

$$\mathcal{T}_Y = \{U \cap Y : U \in \mathcal{T}\}.$$

With this topology, Y is a *subspace* of X.

Exercise 3.3. Check that $\mathcal{T}_Y$ is a topology. Show that if $\mathcal{B}$ is a basis for X, then

$$\mathcal{B}_Y = \{B \cap Y : B \in \mathcal{B}\}$$

is a basis for Y.

Solution. $\varnothing = \varnothing \cap Y$ and $Y = X \cap Y$ belong. The identities

$$\bigcup_\alpha (U_\alpha \cap Y) = \left(\bigcup_\alpha U_\alpha\right) \cap Y, \qquad (U \cap Y) \cap (V \cap Y) = (U \cap V) \cap Y$$

prove closure under arbitrary unions and finite intersections. For the basis assertion, let $y \in U \cap Y$ with U open in X. Choose $B \in \mathcal{B}$ with $y \in B \subseteq U$. Then $y \in B \cap Y \subseteq U \cap Y$. Each $B \cap Y$ is subspace-open, so the basis test applies.

Example 3.7. Let $X = \mathbb{R}$ and $Y = [0, 1]$. The set $(\frac{1}{2}, 1]$ is open in Y because

$$(\tfrac{1}{2}, 1] = (\tfrac{1}{2}, 2) \cap [0, 1],$$

but it is not open in $\mathbb{R}$. Thus openness always refers to an ambient space. The subspace topology of $[0, 1]$ agrees with its order topology: intersecting usual open intervals with $[0, 1]$ gives interior intervals and the endpoint neighborhoods $[0, b)$ and $(a, 1]$, which form an order basis. The same comparison holds for $[0, 1)$ and $(0, 1]$.

Exercise 3.4. If $\mathcal{S}$ is a subbasis for X, show that $\{S \cap Y : S \in \mathcal{S}\}$ is a subbasis for Y.

Solution. A finite intersection of restricted subbasic sets is

$$(S_1 \cap Y) \cap \cdots \cap (S_n \cap Y) = (S_1 \cap \cdots \cap S_n) \cap Y.$$

The finite intersections of members of $\mathcal{S}$ form a basis for X. Restricting this basis gives a basis for Y by the preceding exercise, which proves the claim. The empty intersection is Y on the restricted side, as required.

Theorem 3.8. *Let $A \subseteq X$ and $B \subseteq Y$ be subspaces. The product topology on $A \times B$ agrees with the subspace topology induced from $X \times Y$.*

Exercise 3.5. Prove this agreement by comparing bases.

Solution. An open rectangle in the product of the subspaces has the form

$$(U \cap A) \times (V \cap B),$$

where U and V are open in X and Y. But

$$(U \cap A) \times (V \cap B) = (U \times V) \cap (A \times B).$$

The sets on the right form a basis for the subspace topology inherited from the rectangle basis of $X \times Y$. Thus the two topologies have the same basis and are equal.

Chapter 4

Closed Sets and Limit Points

4.1 Closed sets and subspaces

Recall that a subset A of X is closed if $X \setminus A$ is open.

Example 4.1. In $\mathbb{R}$ with the standard topology, $[0,1]$ is closed because its complement is $(-\infty,0) \cup (1,\infty)$. The set $[0,1)$ is neither open nor closed. It is not open at 0, and its complement $(-\infty,0) \cup [1,\infty)$ is not open at 1. In a discrete space every subset is both open and closed. In the finite complement topology, the closed subsets are precisely X and the finite subsets of X.

Theorem 4.2. *In a topological space X:*

(1) $\varnothing$ and X are closed;

(2) an arbitrary intersection of closed sets is closed;

(3) a finite union of closed sets is closed.

Exercise 4.1. Prove these assertions using complements.

Solution. The complements of $\varnothing$ and X are X and $\varnothing$, which are open. If each A_i is closed, then

$$X \setminus \bigcap_{i \in I} A_i = \bigcup_{i \in I} (X \setminus A_i)$$

is open, since arbitrary unions of open sets are open. Similarly, for closed sets $A_1, \ldots, A_n$,

$$X \setminus \bigcup_{i=1}^{n} A_i = \bigcap_{i=1}^{n} (X \setminus A_i)$$

is open. The conclusions follow from the definition of closedness.

Proposition 4.3. *Let $A \subseteq Y \subseteq X$, where Y has the subspace topology. Then A is closed in Y if and only if there is a closed set C in X such that $A = C \cap Y$.*

Proof. If C is closed in X, then

$$Y \setminus (C \cap Y) = (X \setminus C) \cap Y$$

is open in Y. Hence $C \cap Y$ is closed in Y. Conversely, if A is closed in Y, write

$$Y \setminus A = U \cap Y$$

for an open subset U of X. Let $C = X \setminus U$, which is closed in X. Then

$$A = Y \setminus (U \cap Y) = Y \cap (X \setminus U) = Y \cap C.$$

For example, an element of A belongs to Y and not to U; an element of $Y \cap (X \setminus U)$ cannot belong to $Y \setminus A$, and so must belong to A.

Exercise 4.2. Suppose A is closed in Y and Y is closed in X. Show that A is closed in X.

Solution. By the proposition, $A = C \cap Y$ for some closed subset C of X. Since Y is also closed in X, the intersection $C \cap Y$ is closed in X. Equivalently, if $Y \setminus A = U_1 \cap Y$ with U_1 open in X, and $X \setminus Y = U_2$ is open in X, then

$$X \setminus A = (U_1 \cap Y) \cup U_2 = U_1 \cup U_2,$$

which is open in X.

Exercise 4.3. Give $X = [0,1] \cup (2,3)$ the subspace topology from $\mathbb{R}$. Show that both $[0,1]$ and $(2,3)$ are open and closed in X. Compare closedness in X with closedness in $\mathbb{R}$.

Solution. We have

$$[0,1] = X \cap (-1, 3/2), \qquad (2,3) = X \cap (3/2, 4),$$

so both sets are open in X. They are complements of one another in X, so both are closed in X. The set $[0,1]$ is closed in $\mathbb{R}$, whereas $(2,3)$ is not: every open interval about 2 meets $(2,3)$, but 2 does not belong to that set. Thus closedness must be understood relative to the ambient space.

4.2 Interior and closure

Definition 4.4. For $A \subseteq X$, the *interior* and *closure* of A are

$$\operatorname{Int}_X A = \bigcup \{U \subseteq A : U \text{ is open in } X\}, \qquad \overline{A}^X = \bigcap \{C \supseteq A : C \text{ is closed in } X\}.$$

When the ambient space is clear, write $\operatorname{Int} A$ and $\overline{A}$.

The interior is the largest open subset of A, and the closure is the smallest closed subset of X containing A. In particular,

$$\operatorname{Int} A \subseteq A \subseteq \overline{A}.$$

A set is open if and only if $A = \operatorname{Int} A$, and closed if and only if $A = \overline{A}$.

Definition 4.5. A *neighborhood* of x will mean an open subset of X containing x.

Theorem 4.6 (Neighborhood criterion for closure)**.** *For $A \subseteq X$ and $x \in X$,*

$$x \in \overline{A} \quad \Longleftrightarrow \quad U \cap A \neq \varnothing \text{ for every neighborhood } U \text{ of } x.$$

Exercise 4.4. Prove the neighborhood criterion. Show that it suffices to check basic neighborhoods from any basis of X.

Solution. If $x \notin \overline{A}$, then $U = X \setminus \overline{A}$ is an open set containing x and disjoint from A. Conversely, if an open U contains x and misses A, then $X \setminus U$ is a closed set containing A. Thus $\overline{A} \subseteq X \setminus U$, so $x \notin \overline{A}$. This proves the equivalence by contraposition. If every basic neighborhood of x meets A, any neighborhood U contains a basis element B with $x \in B \subseteq U$, and hence U meets A. The reverse implication is immediate because basic neighborhoods are neighborhoods.

Exercise 4.5. Compute the interior and closure in $\mathbb{R}$ of

$$A = \left\{ \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\}.$$

Solution. Every open interval about 0 contains $1/n$ for sufficiently large n, so $0 \in \overline{A}$. Each point of A belongs to its closure. If $x < 0$ or $x > 1/2$, an interval about x can be chosen disjoint from A. If $0 < x \leq 1/2$ and $x \notin A$, choose N with $1/N < x/2$. The terms with $n \geq N$ lie below $x/2$, while only finitely many terms remain. Since none of those finitely many terms equals x, a sufficiently small interval about x avoids them and the tail. Therefore

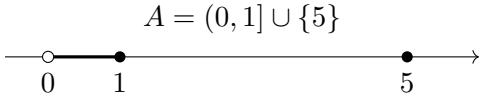
$$\overline{A} = A \cup \{0\}.$$

No nonempty open interval is contained in A, since every interval contains irrational points. Hence $\operatorname{Int} A = \varnothing$.

Example 4.7. For $A = (0, 1] \cup \{5\} \subseteq \mathbb{R}$,

$$\operatorname{Int} A = (0, 1), \qquad \overline{A} = [0, 1] \cup \{5\}.$$

Every interval about 0 meets $(0, 1]$, while 5 remains in the closure because it belongs to A. Neither 1 nor 5 is an interior point.



4.3 Limit points

Definition 4.8. A point $x \in X$ is a *limit point* of $A \subseteq X$ if every neighborhood U of x meets A at a point other than x:

$$U \cap (A \setminus \{x\}) \neq \varnothing.$$

The set of limit points is denoted by A'.

Theorem 4.9. *For every subset A of a topological space,*

$$\overline{A} = A \cup A'.$$

Proof. Certainly $A \subseteq \overline{A}$. If $x \in A'$, every neighborhood of x meets $A \setminus \{x\}$ and therefore meets A. Hence $A' \subseteq \overline{A}$. Conversely, suppose $x \in \overline{A}$ and $x \notin A$. For every neighborhood U of x, the closure criterion gives $U \cap A \neq \varnothing$. Since $x \notin A$, this is the same as $U \cap (A \setminus \{x\}) \neq \varnothing$. Thus $x \in A'$ and $\overline{A} \subseteq A \cup A'$.

Corollary 4.10. *A subset A is closed if and only if it contains all its limit points.*

Proof. The condition $A' \subseteq A$ is equivalent to $A \cup A' = A$, and the theorem identifies $A \cup A'$ with $\overline{A}$.

Exercise 4.6. Determine the limit points of the following subsets in the standard topology:

$$\{1/n : n \geq 2\} \subseteq \mathbb{R}, \qquad \mathbb{Q} \subseteq \mathbb{R}, \qquad D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}.$$

Solution. For the reciprocal sequence, 0 is a limit point, since every interval about 0 contains a positive term $1/n$. Each $1/n$ is isolated from the other terms by choosing an interval smaller than its distances to the neighboring terms. The preceding closure calculation excludes every other point. Thus its set of limit points is $\{0\}$.

For every $x \in \mathbb{R}$ and every interval about x, there is a rational number different from x in that interval. For example, choose a smaller nonempty subinterval that does not contain x, then use the density of $\mathbb{Q}$. Consequently $\mathbb{Q}' = \mathbb{R}$ and $\overline{\mathbb{Q}} = \mathbb{R}$.

For D, every point p with $\|p\| < 1$ has nearby distinct points of D. For $\|p\| = 1$, the points $(1 - 1/n)p$ belong to D and tend to p, so each neighborhood of p meets D. If $\|p\| > 1$, the ball of radius $(\|p\| - 1)/2$ about p misses D, by the triangle inequality. Therefore

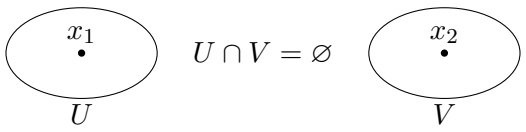
$$D' = \overline{D} = \{(x, y) : x^2 + y^2 \leq 1\}.$$

In particular, every point of the open disk is itself a limit point.

4.4 Hausdorff spaces and convergence

Definition 4.11. A sequence $(x_n)_{n \in \mathbb{N}}$ in a topological space X *converges* to $a \in X$ if, for every neighborhood U of a, there is $N \in \mathbb{N}$ such that $x_n \in U$ whenever $n \geq N$.

Definition 4.12. A space X is *Hausdorff* if, for any distinct $x_1, x_2 \in X$, there are disjoint open sets U, V such that $x_1 \in U$ and $x_2 \in V$.

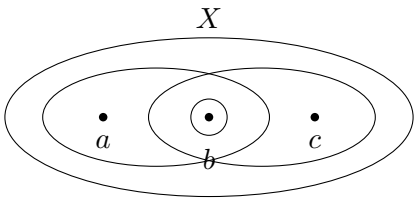


Example 4.13. The real line is Hausdorff. For $a < b$, let $m = (a + b)/2$. The open sets $(-\infty, m)$ and (m, ∞) are disjoint neighborhoods of a and b. Singletons such as $\{0\} \subseteq \mathbb{R}$ and $\{(0, 0)\} \subseteq \mathbb{R}^2$ are closed.

Example 4.14. Let

$$X = \{a, b, c\}, \qquad \mathcal{T} = \{\varnothing, \{b\}, \{a, b\}, \{b, c\}, X\}.$$

The two proper two-point open sets overlap at b, and their intersection $\{b\}$ is open.



The singleton $\{b\}$ is not closed, since its complement $\{a, c\}$ is not open. The space is not Hausdorff: every neighborhood of a contains b, so no neighborhood of a can be disjoint from a neighborhood of b. The constant sequence $x_n = b$ converges to all three points. Indeed, every nonempty open set contains b.

Lemma 4.15. *Every finite subset of a Hausdorff space is closed.*

Proof. It suffices to prove that every singleton $\{a\}$ is closed, since a finite union of closed sets is closed. For each $b \in X \setminus \{a\}$, choose disjoint neighborhoods U_b of b and V_b of a. Then $U_b \subseteq X \setminus \{a\}$, and

$$X \setminus \{a\} = \bigcup_{b \neq a} U_b$$

is open. Thus $\{a\}$ is closed.

Definition 4.16. A space X satisfies the T_1 *condition* if every singleton is closed. Equivalently, every finite subset is closed.

Every Hausdorff space is T_1.

Exercise 4.7. In a T_1 space, show that x is a limit point of A if and only if every neighborhood of x contains infinitely many points of A.

Solution. If every neighborhood contains infinitely many points of A, removing x leaves a nonempty intersection, so $x \in A'$. Conversely, suppose $x \in A'$ and some neighborhood U meets A in only finitely many points. The set $F = (U \cap A) \setminus \{x\}$ is finite and therefore closed. Then $U \setminus F$ is an open neighborhood of x and

$$(U \setminus F) \cap (A \setminus \{x\}) = \varnothing,$$

contrary to $x \in A'$. Hence every neighborhood contains infinitely many points of A.

Exercise 4.8. Show that a sequence in a Hausdorff space has at most one limit.

Solution. Suppose $x_n \to a$ and $x_n \to b$ with $a \neq b$. Choose disjoint neighborhoods U of a and V of b. There are integers N_1, N_2 such that $x_n \in U$ for $n \geq N_1$ and $x_n \in V$ for $n \geq N_2$. For $n \geq \max\{N_1, N_2\}$ this gives $x_n \in U \cap V = \varnothing$, a contradiction.

Chapter 5

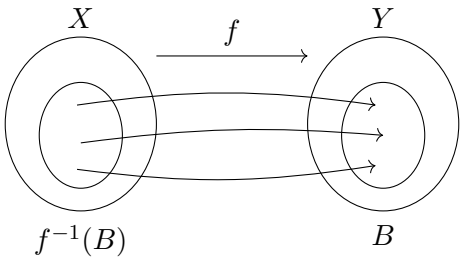
Continuous Functions and Homeomorphisms

5.1 Inverse images and the definition of continuity

Recall that for a function $f : X \to Y$ and a subset $B \subseteq Y$, the *preimage*, or *inverse image*, is

$$f^{-1}(B) = \{x \in X : f(x) \in B\}.$$

This notation does not require f to be invertible.



Exercise 5.1. Let $f : X \to Y$, let $\{B_\alpha\}_{\alpha \in J}$ be subsets of Y, let $B \subseteq Y$, and let $A \subseteq X$. Prove

$$\begin{aligned} f^{-1}\left(\bigcup_{\alpha \in J} B_\alpha\right) &= \bigcup_{\alpha \in J} f^{-1}(B_\alpha), \\ f^{-1}\left(\bigcap_{\alpha \in J} B_\alpha\right) &= \bigcap_{\alpha \in J} f^{-1}(B_\alpha), \\ f^{-1}(Y \setminus B) &= X \setminus f^{-1}(B), \\ A \subseteq f^{-1}(B) \quad &\iff \quad f(A) \subseteq B. \end{aligned}$$

Solution. A point x belongs to the inverse image of $\bigcup_\alpha B_\alpha$ exactly when $f(x)$ belongs to at least one B_α, or equivalently when x belongs to at least one $f^{-1}(B_\alpha)$. This proves the union identity. Replacing "at least one" by "every" proves the intersection identity. For the empty index set, the intersection in Y is Y, its inverse image is X, and the intersection in X is also X. Moreover,

$$x \in f^{-1}(Y \setminus B) \iff f(x) \notin B \iff x \notin f^{-1}(B),$$

which proves the complement identity. Finally, $A \subseteq f^{-1}(B)$ says that $f(x) \in B$ for every $x \in A$, precisely the assertion $f(A) \subseteq B$.

Definition 5.1. Let X and Y be topological spaces. A function $f : X \to Y$ is *continuous* if $f^{-1}(V)$ is open in X for every open subset V of Y.

Proposition 5.2. *If $\mathcal{B}_Y$ is a basis for the topology on Y, a function $f : X \to Y$ is continuous if and only if $f^{-1}(B)$ is open in X for every $B \in \mathcal{B}_Y$.*

Proof. Continuity immediately gives the condition because basis elements are open. Conversely, write any open subset V of Y as a union $V = \bigcup_\alpha B_\alpha$ of basis elements. Then

$$f^{-1}(V) = \bigcup_\alpha f^{-1}(B_\alpha),$$

which is open in X if every $f^{-1}(B_\alpha)$ is open.

5.1.1 The calculus definition

For $x \in \mathbb{R}$ and $\delta > 0$, write

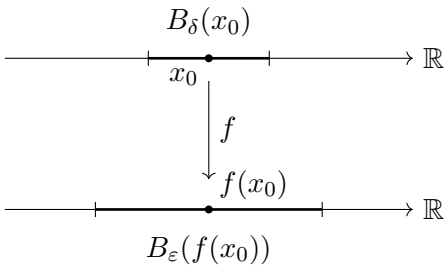
$$B_\delta(x) = (x - \delta, x + \delta).$$

These intervals form a basis for the standard topology on $\mathbb{R}$. The calculus condition for continuity of $f : \mathbb{R} \to \mathbb{R}$ at x_0 is

$$\text{for every } \varepsilon > 0 \text{ there is } \delta > 0 \text{ such that } |x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon.$$

Equivalently,

$$f(B_\delta(x_0)) \subseteq B_\varepsilon(f(x_0)), \qquad \text{or} \qquad B_\delta(x_0) \subseteq f^{-1}(B_\varepsilon(f(x_0))).$$



Exercise 5.2. Show that the calculus definition of continuity for $f : \mathbb{R} \to \mathbb{R}$ agrees with the topological definition for the standard topologies.

Solution. Suppose the calculus condition holds at every x_0. Let V be open and $x_0 \in f^{-1}(V)$. Since $f(x_0) \in V$, choose $\varepsilon > 0$ with $B_\varepsilon(f(x_0)) \subseteq V$. Choose $\delta > 0$ from continuity at x_0. Then

$$x_0 \in B_\delta(x_0) \subseteq f^{-1}(B_\varepsilon(f(x_0))) \subseteq f^{-1}(V).$$

Thus every point of $f^{-1}(V)$ has an open interval contained in that inverse image, so $f^{-1}(V)$ is open. Conversely, suppose f is topologically continuous. Given x_0 and $\varepsilon > 0$, the set $f^{-1}(B_\varepsilon(f(x_0)))$ is open and contains x_0. Hence it contains $B_\delta(x_0)$ for some $\delta > 0$. This is exactly the calculus condition.

Note 5.3. With the standard product topologies on $\mathbb{R}^n$ and $\mathbb{R}^m$, topological continuity of $f : \mathbb{R}^n \to \mathbb{R}^m$ likewise agrees with the usual calculus definition. Euclidean open balls and open products of intervals generate the same topology: a sufficiently small box about a point lies in a given ball, and a sufficiently small ball lies in a given box.

5.2 Equivalent criteria for continuity

Theorem 5.4. *Let $f : X \to Y$ be a function between topological spaces. The following conditions are equivalent:*

(1) f is continuous.

(2) For every subset $A \subseteq X$,
$$f(\overline{A}) \subseteq \overline{f(A)}.$$

(3) For every closed subset $B \subseteq Y$, the inverse image $f^{-1}(B)$ is closed in X.

(4) For every $x \in X$ and every neighborhood V of $f(x)$, there is a neighborhood U of x such that $f(U) \subseteq V$.

Exercise 5.3. Prove the equivalences by showing

$$(1) \Longrightarrow (2) \Longrightarrow (3) \Longrightarrow (1), \qquad (1) \Longrightarrow (4) \Longrightarrow (1).$$

Solution. (1) $\Rightarrow$ (2): Let $x \in \overline{A}$. If V is any neighborhood of $f(x)$, then $f^{-1}(V)$ is an open neighborhood of x. Hence $f^{-1}(V) \cap A \neq \varnothing$. For a point a in this intersection, $f(a) \in V \cap f(A)$. Thus every neighborhood of $f(x)$ meets $f(A)$, so $f(x) \in \overline{f(A)}$.

(2) $\Rightarrow$ (3): Let B be closed in Y and put $A = f^{-1}(B)$. Then $f(A) \subseteq B$, and therefore

$$f(\overline{A}) \subseteq \overline{f(A)} \subseteq B.$$

The last inclusion holds because B is a closed set containing $f(A)$. It follows that $\overline{A} \subseteq f^{-1}(B) = A$. Since $A \subseteq \overline{A}$, we have equality, so A is closed.

(3) $\Rightarrow$ (1): For an open set $V \subseteq Y$, its complement $Y \setminus V$ is closed. Thus

$$f^{-1}(V) = X \setminus f^{-1}(Y \setminus V)$$

is open, proving continuity.

(1) $\Rightarrow$ (4): Given x and V, take $U = f^{-1}(V)$. It is an open neighborhood of x and satisfies $f(U) \subseteq V$.

(4) $\Rightarrow$ (1): Let V be open in Y. For each $x \in f^{-1}(V)$, condition (4) gives an open neighborhood U_x with $f(U_x) \subseteq V$, hence $U_x \subseteq f^{-1}(V)$. Therefore

$$f^{-1}(V) = \bigcup_{x \in f^{-1}(V)} U_x$$

is open in X.

Example 5.5. Every function from a discrete space X to any topological space Y is continuous: the inverse image of an open subset of Y is a subset of X, and every subset of X is open.

Example 5.6. Continuity depends on the topologies. Let $\mathbb{R}_\ell$ denote the real line with the lower limit topology. The identity function

$$f : \mathbb{R} \longrightarrow \mathbb{R}_\ell, \qquad f(x) = x,$$

is not continuous, since the basic open set $[0, 1)$ in $\mathbb{R}_\ell$ has inverse image $[0, 1)$, which is not open in the standard topology. In the other direction,

$$g : \mathbb{R}_\ell \longrightarrow \mathbb{R}, \qquad g(x) = x,$$

is continuous. Indeed,

$$g^{-1}((a, b)) = (a, b) = \bigcup_{a<x<b} [x, b)$$

is open in $\mathbb{R}_\ell$. It suffices to check standard open intervals, since they form a basis for $\mathbb{R}$.

5.3 Homeomorphisms

Definition 5.7. A function $f : X \to Y$ is a *homeomorphism* if it is continuous and there is a continuous function $g : Y \to X$ satisfying

$$f \circ g = \mathrm{id}_Y, \qquad g \circ f = \mathrm{id}_X .$$

The spaces X and Y are *homeomorphic* if such a map exists.

Exercise 5.4. Show that this definition is equivalent to saying that f is a continuous bijection and its inverse $f^{-1} : Y \to X$ is continuous.

Solution. If $g \circ f = \mathrm{id}_X$ and $f(x_1) = f(x_2)$, applying g gives $x_1 = x_2$, so f is injective. For each $y \in Y$, the identity $f(g(y)) = y$ proves surjectivity. Thus f is a bijection and the two identities show $g = f^{-1}$. Conversely, if f is a continuous bijection with continuous inverse, take $g = f^{-1}$; the two required identities follow from the definition of an inverse function.

Proposition 5.8. *A bijection $f : X \to Y$ is a homeomorphism if and only if*

$$U \text{ is open in } X \quad \Longleftrightarrow \quad f(U) \text{ is open in } Y \qquad (U \subseteq X).$$

Proof. If f is a homeomorphism and U is open in X, then

$$f(U) = (f^{-1})^{-1}(U)$$

is open in Y, since f^{-1} is continuous. If $f(U)$ is open in Y, then $U = f^{-1}(f(U))$ is open by continuity of f. Conversely, suppose the displayed equivalence holds. For an open set $V \subseteq Y$, set $U = f^{-1}(V)$. Bijectivity gives $f(U) = V$, so the equivalence makes U open. Thus f is continuous. For an open $U \subseteq X$, the same equivalence says that $(f^{-1})^{-1}(U) = f(U)$ is open in Y, so f^{-1} is continuous as well.

Example 5.9. With the standard subspace topologies, the following are homeomorphisms:

$$\begin{aligned} f : [0,1] &\longrightarrow [5,7], & f(x) &= 2x+5, & f^{-1}(y) &= \frac{y-5}{2}, \\ h : \mathbb{R} &\longrightarrow \mathbb{R}, & h(x) &= 2x+1, & h^{-1}(y) &= \frac{y-1}{2}, \\ k : (0,1) &\longrightarrow (0,2), & k(x) &= 2x, & k^{-1}(y) &= \frac{y}{2}. \end{aligned}$$

Each displayed map and inverse is continuous. Substitution in either order gives the appropriate identity function; for instance,

$$\frac{(2x+5)-5}{2} = x, \qquad 2\left(\frac{y-5}{2}\right)+5 = y.$$

Exercise 5.5. Show that $(-1,1)$ is homeomorphic to $\mathbb{R}$ by the function

$$f(x) = \frac{x}{1-x^2}.$$

Solution. The denominator is positive on $(-1,1)$, so f is continuous there. Also,

$$f'(x) = \frac{1+x^2}{(1-x^2)^2} > 0.$$

Thus f is strictly increasing and injective. Its endpoint limits are

$$\lim_{x\to -1^+} f(x) = -\infty, \qquad \lim_{x\to 1^-} f(x) = +\infty.$$

The intermediate value theorem gives surjectivity onto $\mathbb{R}$. Solving $y = x/(1-x^2)$ yields an inverse, written without a removable singularity as

$$g(y) = \frac{2y}{1+\sqrt{1+4y^2}}.$$

Put $s = \sqrt{1+4y^2}$. Since $|2y| < 1+s$, the value $g(y)$ lies in $(-1,1)$, and

$$1 - g(y)^2 = 1 - \frac{4y^2}{(1+s)^2} = \frac{(1+s)^2-(s^2-1)}{(1+s)^2} = \frac{2}{1+s}.$$

Consequently $f(g(y)) = y$ for every $y \in \mathbb{R}$. Since f is injective, this also gives $g(f(x)) = x$. The formula for g is continuous on $\mathbb{R}$, its denominator being at least 2. Thus f is a homeomorphism.

5.3.1 Bijections and continuity

Example 5.10. Let $Y = [0,1] \cup (2,3]$ have the subspace topology from $\mathbb{R}$. Define a bijection

$$f : [0,2] \longrightarrow Y, \qquad f(x) = \begin{cases} x, & 0 \le x \le 1, \\ x+1, & 1 < x \le 2. \end{cases}$$

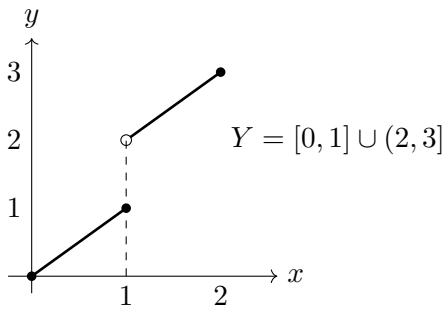
It is not continuous. The set

$$V = (1/2, 3/2) \cap Y = (1/2, 1]$$

is open in Y, but

$$f^{-1}(V) = (1/2, 1]$$

is not open in $[0,2]$: every relative neighborhood of 1 also contains points greater than 1.



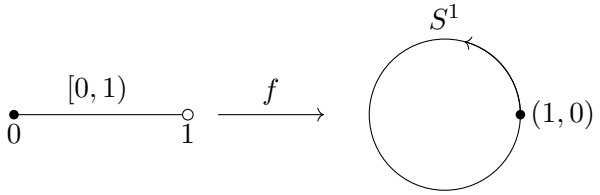
Example 5.11. A continuous bijection need not be a homeomorphism. Let

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$$

with the subspace topology, and define

$$f : [0, 1) \longrightarrow S^1, \qquad f(t) = (\cos 2\pi t, \sin 2\pi t).$$

The map traverses the circle once, starting at $(1, 0)$ and omitting the second occurrence of that point at $t = 1$.



Exercise 5.6. Verify that the circle map f is a continuous bijection, and show that its inverse is not continuous.

Solution. For $s, t \in [0, 1)$, the trigonometric identities give

$$\|f(s) - f(t)\|^2 = 2 - 2\cos(2\pi(s - t)) = 4\sin^2(\pi(s - t)),$$

so $\|f(s) - f(t)\| \leq 2\pi|s - t|$. Thus f is continuous into $\mathbb{R}^2$. If $V = W \cap S^1$ is open in S^1, with W open in $\mathbb{R}^2$, then $f^{-1}(V) = f^{-1}(W)$ because the image lies in S^1. This also proves continuity into the subspace S^1. Every point on the unit circle has an angle $\theta \in [0, 2\pi)$, proving surjectivity. If $f(s) = f(t)$, the angles differ by an integer multiple of 2π, so $s - t \in \mathbb{Z}$. For $s, t \in [0, 1)$, their difference lies in $(-1, 1)$; the only possible integer is 0. Thus $s = t$, and f is bijective.

The set $U = [0, 1/2)$ is open in $[0, 1)$, since $U = (-1, 1/2) \cap [0, 1)$. Its image is

$$f(U) = \{(x, y) \in S^1 : y > 0\} \cup \{(1, 0)\}.$$

This set is not open in S^1. Every relative neighborhood of $(1, 0)$ contains points on the lower semicircle, for example $f(1 - 1/n)$ for sufficiently large n, whereas those points do not belong to $f(U)$. Therefore $(f^{-1})^{-1}(U) = f(U)$ is not open. The inverse map $f^{-1} : S^1 \to [0, 1)$ is not continuous.

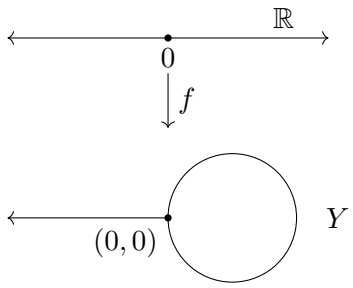
Example 5.12 (A line and a loop)**.** Let

$$Y = \big((-\infty, 0] \times \{0\}\big) \cup \{(x, y) \in \mathbb{R}^2 : (x-1)^2 + y^2 = 1\}$$

carry the subspace topology from $\mathbb{R}^2$. Thus a leftward ray is attached to a circle at $(0, 0)$. Define $f : \mathbb{R} \to Y$ by

$$f(t) = \begin{cases} (t, 0), & t \le 0, \\ (1 - \cos\theta(t), \sin\theta(t)), & t > 0, \end{cases} \qquad \theta(t) = \frac{2\pi t}{1+t} \quad (t > 0).$$

The function is a continuous bijection, but its inverse is not continuous.



Exercise 5.7. Verify the assertions in Example 5.12, and examine the image of the open interval $(-1, 1)$.

Solution. Both formulas are continuous on their respective sides of 0. As $t \to 0^+$, $\theta(t) \to 0$, so the second formula tends to $(0, 0) = f(0)$; the first formula has the same limit from the left. Thus f is continuous into $\mathbb{R}^2$, and hence into its subspace Y.

For $t \le 0$, the map is a bijection onto the ray. For $t > 0$, the angle $\theta(t)$ increases bijectively from 0 to 2π, with neither endpoint attained. It therefore traces the circle minus $(0, 0)$ exactly once. The two images are disjoint, and together they equal Y, so f is bijective.

Let $W = (-1, 1)$. Its image contains $(0, 0)$ but omits $f(n)$ for every positive integer n. Nevertheless,

$$\|f(n)\| = 2\sin\left(\frac{\pi}{n+1}\right) \le \frac{2\pi}{n+1} \longrightarrow 0.$$

Every relative neighborhood of $(0, 0)$ in Y thus contains points $f(n)$ outside $f(W)$. Consequently $f(W)$ is not open in Y. Since

$$(f^{-1})^{-1}(W) = f(W),$$

the inverse function $f^{-1} : Y \to \mathbb{R}$ is not continuous.

5.4 Constructing continuous functions

5.4.1 Constant maps and inclusions

Exercise 5.8. Show that every constant map $f : X \to Y$, $f(x) = c$ for a fixed $c \in Y$, is continuous.

Solution. For any open subset V of Y,

$$f^{-1}(V) = \begin{cases} X, & c \in V, \\ \varnothing, & c \notin V. \end{cases}$$

Both possible inverse images are open in X.

Exercise 5.9. Let $A \subseteq X$ have the subspace topology. Show that the inclusion $i : A \to X$, $i(a) = a$, is continuous.

Solution. For an open subset U of X, the inverse image is $i^{-1}(U) = A \cap U$, which is open in A by the definition of the subspace topology.

5.4.2 Compositions and restrictions

Proposition 5.13. *If $g : X \to Y$ and $f : Y \to Z$ are continuous, then the composite $f \circ g : X \to Z$ is continuous.*

Proof. For an open subset $W \subseteq Z$,

$$\begin{aligned} (f \circ g)^{-1}(W) &= \{x \in X : f(g(x)) \in W\} \\ &= \{x \in X : g(x) \in f^{-1}(W)\} \\ &= g^{-1}(f^{-1}(W)). \end{aligned}$$

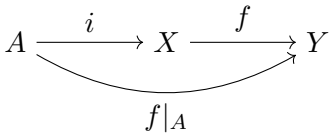
The set $f^{-1}(W)$ is open in Y by continuity of f. Its inverse image under g is open in X by continuity of g.

Proposition 5.14 (Restricting the domain)**.** *Let $f : X \to Y$ be continuous, and let $A \subseteq X$ have the subspace topology. Then the restriction*

$$f|_A : A \to Y, \qquad f|_A(a) = f(a),$$

is continuous.

Proof. If $i : A \to X$ is the inclusion, then $f|_A = f \circ i$. Both factors are continuous, so the composition is continuous. Equivalently, $(f|_A)^{-1}(V) = A \cap f^{-1}(V)$ is open in A whenever V is open in Y.



Example 5.15. The map $F : \mathbb{R}^2 \to \mathbb{R}^2$ given by $F(x, y) = (4x, 4y)$ is continuous, so its restriction $F|_{S^1} : S^1 \to \mathbb{R}^2$ is continuous. The domain carries the subspace topology from $\mathbb{R}^2$.

5.4.3 Restricting the range

Proposition 5.16. *Suppose $f : X \to Y$ is continuous and $f(X) \subseteq Z \subseteq Y$. Give Z the subspace topology. Then*

$$g : X \to Z, \qquad g(x) = f(x),$$

is continuous. In particular, this holds for $Z = f(X)$.

Proof. Every open subset of Z has the form $Z \cap V$, where V is open in Y. Since $f(x) \in Z$ for every $x \in X$,

$$g^{-1}(Z \cap V) = f^{-1}(V),$$

which is open in X.

5.4.4 The pasting lemma

Theorem 5.17 (Pasting lemma)**.** *Suppose $X = A \cup B$, where A and B are closed subsets of X. Let $f : A \to Y$ and $g : B \to Y$ be continuous maps that agree on $A \cap B$. Then*

$$h : X \to Y, \qquad h(x) = \begin{cases} f(x), & x \in A, \\ g(x), & x \in B, \end{cases}$$

is a well-defined continuous function.

Exercise 5.10. Prove the pasting lemma using inverse images of closed sets.

Solution. Every $x \in X$ belongs to A or B. If it belongs to both, $f(x) = g(x)$, so the formula for h is well-defined. Let C be closed in Y. By continuity, $f^{-1}(C)$ is closed in A, and $g^{-1}(C)$ is closed in B. Since A and B are closed in X, these inverse images are also closed in X. Now

$$h^{-1}(C) = f^{-1}(C) \cup g^{-1}(C)$$

is a union of two closed subsets of X, hence is closed. The closed-set criterion for continuity proves that h is continuous.

Chapter 6

Products and Uniform Topologies

6.1 Projections and component functions

Recall that $X \times Y$ has the product topology. Its projections are

$$\pi_1 : X \times Y \to X, \quad \pi_1(x, y) = x, \qquad \pi_2 : X \times Y \to Y, \quad \pi_2(x, y) = y.$$

For open subsets $U \subseteq X$ and $V \subseteq Y$,

$$\pi_1^{-1}(U) = U \times Y, \qquad \pi_2^{-1}(V) = X \times V.$$

Both are open in the product topology, so the projections are continuous. These sets form a subbasis, and their pairwise intersections give the basic open rectangles:

$$(U \times Y) \cap (X \times V) = U \times V.$$

Theorem 6.1. *Let A, X, Y be topological spaces and let*

$$f : A \to X \times Y, \qquad f(a) = (f_1(a), f_2(a)).$$

Then f is continuous if and only if both component functions $f_1 : A \to X$ and $f_2 : A \to Y$ are continuous.

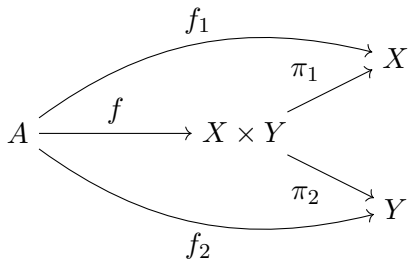
Proof. If f is continuous, then

$$f_1 = \pi_1 \circ f, \qquad f_2 = \pi_2 \circ f$$

are continuous, since compositions of continuous maps are continuous. Conversely, suppose both component functions are continuous. It suffices to consider a basic open rectangle $U \times V$. Its inverse image is

$$\begin{aligned} f^{-1}(U \times V) &= \{a \in A : f(a) \in U \times V\} \\ &= \{a \in A : f_1(a) \in U \text{ and } f_2(a) \in V\} \\ &= f_1^{-1}(U) \cap f_2^{-1}(V). \end{aligned}$$

Each factor is open in A, so their intersection is open. The basis criterion proves that f is continuous.



Example 6.2. The function

$$f : \mathbb{R} \to \mathbb{R}^2, \qquad f(t) = (\cos t, \sin t),$$

is continuous because its two component functions are continuous. Since its image lies in S^1, restricting the range gives a continuous map $\mathbb{R} \to S^1$ as well.

6.2 Indexed tuples and products of sets

Definition 6.3. Let J be an index set. A *J-tuple of elements of X* is a function $x : J \to X$. For $\alpha \in J$, write $x_\alpha = x(\alpha)$ and denote the tuple by $(x_\alpha)_{\alpha \in J}$. The set of all such functions is denoted by X^J.

Example 6.4. A point (x_1, x_2, x_3) of $\mathbb{R}^3$ is a function from $\{1, 2, 3\}$ to $\mathbb{R}$. Likewise,

$$\prod_{i \in \mathbb{Z}_{>0}} \mathbb{R} = \{(x_1, x_2, x_3, \ldots) : x_i \in \mathbb{R} \text{ for every } i\} = \mathbb{R}^{\mathbb{Z}_{>0}}.$$

Thus an infinite tuple is simply a function assigning a real number to each positive integer.

Exercise 6.1. Let $\{A_\alpha\}_{\alpha \in J}$ be an indexed family of sets. Express their Cartesian product in terms of indexed tuples, and describe its coordinate maps.

Solution. The Cartesian product is

$$\prod_{\alpha \in J} A_\alpha = \left\{ x : J \to \bigcup_{\alpha \in J} A_\alpha : x(\alpha) \in A_\alpha \text{ for every } \alpha \in J \right\}.$$

Writing $x = (x_\alpha)_{\alpha \in J}$, the projection onto coordinate $\beta \in J$ is

$$\pi_\beta : \prod_{\alpha \in J} A_\alpha \to A_\beta, \qquad \pi_\beta(x) = x_\beta.$$

For $J = \{1, 2\}$ this recovers the ordinary Cartesian product and its two projections. If every A_α equals one fixed set X, the product is X^J.

In this chapter and the next, sequences are indexed by $\mathbb{Z}_+ = \{1, 2, 3, \ldots\}$.

6.3 Products indexed by an arbitrary set

Recall that a basic open set in a finite product is a product of open sets, and a subbasic open set restricts one coordinate. For an infinite product, the number of coordinates restricted distinguishes the product topology from the box topology.

Definition 6.5. Let J be an index set and let X_α be a topological space for each $\alpha \in J$. The Cartesian product is

$$X = \prod_{\alpha \in J} X_\alpha = \{(x_\alpha)_{\alpha \in J} : x_\alpha \in X_\alpha \text{ for every } \alpha \in J\}.$$

The coordinate projection is $\pi_\beta : X \to X_\beta$, $\pi_\beta((x_\alpha)) = x_\beta$. The *product topology* is generated by the subbasis

$$\{\pi_\beta^{-1}(U) : \beta \in J,\ U \text{ open in } X_\beta\}.$$

Its basic open sets are $\prod_{\alpha \in J} U_\alpha$, where every U_α is open and $U_\alpha = X_\alpha$ except for finitely many α. The *box topology* has basis all products $\prod_{\alpha \in J} U_\alpha$ of open sets, without the finiteness condition.

Example 6.6. For $J = \mathbb{Z}$ and $X_\alpha = [\alpha, \alpha + 1]$,

$$\prod_{\alpha \in \mathbb{Z}} [\alpha, \alpha + 1] = \{(x_\alpha)_{\alpha \in \mathbb{Z}} : \alpha \le x_\alpha \le \alpha + 1\}.$$

For $J = \mathbb{R}$ and $X_\alpha = \mathbb{R}$ for every α, the product $\mathbb{R}^{\mathbb{R}}$ is the set of all functions $\mathbb{R} \to \mathbb{R}$: the tuple (x_α) corresponds to the function $\alpha \mapsto x_\alpha$.

Proposition 6.7. *A map $F : A \to \prod_{\alpha \in J} X_\alpha$, with the product topology on its codomain, is continuous if and only if each $\pi_\alpha \circ F$ is continuous.*

Exercise 6.2. Prove Proposition 6.7.

Solution. Every projection is continuous because the inverse image of an open set is subbasic open. Thus continuity of F implies continuity of all its coordinates. Conversely,

$$F^{-1}(\pi_\alpha^{-1}(U)) = (\pi_\alpha \circ F)^{-1}(U)$$

is open whenever U is open. Inverse images preserve unions and finite intersections, so inverse images of all product-open sets are open.

6.4 Metrics and induced topologies

Definition 6.8. A *metric* on a set X is a function $d : X \times X \to [0, \infty)$ satisfying, for all $x, y, z \in X$,

(i) $d(x, y) = 0$ if and only if $x = y$;

(ii) $d(x, y) = d(y, x)$;

(iii) $d(x, z) \le d(x, y) + d(y, z)$.

The pair (X, d) is a *metric space*. The open ball about x of radius $r > 0$ is

$$B_d(x, r) = \{y \in X : d(x, y) < r\}.$$

The *metric topology*, or topology induced by d, consists of the sets $U \subseteq X$ such that each $x \in U$ has some $r > 0$ with $B_d(x, r) \subseteq U$. A topological space is *metrizable* if its topology is induced by some metric.

Exercise 6.3. Show that the open balls form a basis for the metric topology.

Solution. Every point lies in a ball. If $z \in B_d(x, r) \cap B_d(y, s)$, let

$$t = \min\{r - d(x, z), s - d(y, z)\} > 0.$$

For $w \in B_d(z, t)$, the triangle inequality gives

$$d(x, w) \le d(x, z) + d(z, w) < d(x, z) + t \le r,$$

and similarly $d(y, w) < s$. Thus $B_d(z, t)$ lies in the intersection, proving the basis intersection condition. The unions of these basic sets are exactly the sets in the definition of the metric topology.

6.5 Three topologies on the real sequence space

Write $\mathbb{R}^\omega = \prod_{i=1}^{\infty} \mathbb{R}$, whose points are real sequences $x = (x_1, x_2, \ldots)$. Define the bounded metric on $\mathbb{R}$ by

$$\bar{d}(s, t) = \min\{|s - t|, 1\}.$$

Definition 6.9. The *uniform metric* on $\mathbb{R}^\omega$ is

$$\bar{\rho}(x, y) = \sup_{i \ge 1} \bar{d}(x_i, y_i) = \sup_{i \ge 1} \min\{|x_i - y_i|, 1\}.$$

The topology it induces is called the *uniform topology*.

Example 6.10. A supremum need not belong to the set:

$$\sup\{q \in \mathbb{Q} : q \ge 0,\ q^2 < 2\} = \sqrt{2}.$$

Every member is less than $\sqrt{2}$, while rational numbers occur arbitrarily close to $\sqrt{2}$ from below.

Exercise 6.4. Verify that $\bar{d}$ and $\bar{\rho}$ are metrics. Describe a ball of radius $0 < \varepsilon < 1$ in the uniform metric, and compare the three topologies on $\mathbb{R}^\omega$.

Solution. The identity

$$\min\{a + b, 1\} \le \min\{a, 1\} + \min\{b, 1\} \qquad (a, b \ge 0)$$

and the usual triangle inequality prove the triangle inequality for $\bar{d}$. Symmetry and positive definiteness are immediate. For $\bar{\rho}$, its supremum exists in $[0, 1]$, and

$$\bar{d}(x_i, z_i) \le \bar{d}(x_i, y_i) + \bar{d}(y_i, z_i) \le \bar{\rho}(x, y) + \bar{\rho}(y, z)$$

for every i. Taking the supremum proves the triangle inequality. If $\bar{\rho}(x,y) = 0$, then $x_i = y_i$ for every i.

For $0 < \varepsilon < 1$,

$$B_{\bar{\rho}}(x,\varepsilon) = \left\{ y : \sup_i |x_i - y_i| < \varepsilon \right\} = \bigcup_{0<r<\varepsilon} \prod_{i=1}^{\infty} (x_i - r, x_i + r).$$

In particular,

$$B_{\bar{\rho}}(x,\varepsilon) \subseteq \prod_i (x_i - \varepsilon, x_i + \varepsilon),$$

but equality need not hold. For example, $y_i = x_i + \varepsilon(1 - 1/(i+1))$ belongs to the product on the right, while $\bar{\rho}(x,y) = \varepsilon$.

Let U be a product-basic neighborhood of x, restricting a finite set F of coordinates. Choose $0 < \delta < 1$ small enough that $(x_i - \delta, x_i + \delta) \subseteq U_i$ for every $i \in F$. Then $B_{\bar{\rho}}(x,\delta) \subseteq U$. Hence every product-open set is uniform-open. Each uniform ball is box-open by the displayed union, so

$$\mathcal{T}_{\text{product}} \subseteq \mathcal{T}_{\text{uniform}} \subseteq \mathcal{T}_{\text{box}}.$$

Both inclusions are strict. The uniform ball $B_{\bar{\rho}}(0, 1/2)$ contains no product-basic neighborhood of 0, since an unrestricted coordinate can be changed to 1. The box-open set

$$V = \prod_{i=1}^{\infty} (-1/i, 1/i)$$

contains no uniform ball about 0: given $r > 0$, choose i so large that $1/i < \min\{r, 1\}/2$, and change just its ith coordinate to $\min\{r, 1\}/2$.

Exercise 6.5. Find a metric giving the product topology on $\mathbb{R}^\omega$.

Solution. Set

$$d_{\mathrm{p}}(x,y) = \sup_{i \geq 1} \frac{\bar{d}(x_i, y_i)}{i}.$$

Positive definiteness, symmetry, and the triangle inequality follow coordinatewise as for $\bar{\rho}$. Given a product-basic neighborhood U of x restricting finitely many coordinates F, choose $0 < \delta_i < 1$ with $(x_i - \delta_i, x_i + \delta_i) \subseteq U_i$ for $i \in F$. Any d_{p}-ball about x with radius at most $\min_{i \in F}(\delta_i / i)$ lies in U; if F is empty, there is no restriction.

Conversely, given $r > 0$, choose an integer $N \geq 1$ with $1/(N+1) < r$. Choose $0 < \delta_i < \min\{1, ir\}$ for $1 \leq i \leq N$. The product-basic neighborhood

$$U = \prod_{i=1}^{N} (x_i - \delta_i, x_i + \delta_i) \times \prod_{i > N} \mathbb{R}$$

is contained in $B_{d_{\mathrm{p}}}(x, r)$, because

$$d_{\mathrm{p}}(x,y) \leq \max\left\{ \max_{1 \leq i \leq N} \frac{|x_i - y_i|}{i}, \frac{1}{N+1} \right\} < r \quad (y \in U).$$

The two topologies therefore agree.

Exercise 6.6. Consider $F : \mathbb{R} \to \mathbb{R}^\omega$, $F(t) = (t, t, t, \ldots)$. Determine whether it is continuous for each of the product, uniform, and box topologies.

Solution. For the product topology, each coordinate $\pi_i \circ F$ is the identity on $\mathbb{R}$, so F is continuous. For the uniform topology,

$$\bar{\rho}(F(s), F(t)) = \min\{|s-t|, 1\},$$

so F is continuous. For the box topology,

$$F^{-1}\left(\prod_{i=1}^{\infty}(-1/i, 1/i)\right) = \{0\},$$

which is not open in $\mathbb{R}$. Thus F is not continuous in the box topology.

6.6 Subsets of the sequence space

Exercise 6.7. In $\mathbb{R}^\omega$, consider

$$\begin{aligned} A &= \{x : x_i = 0 \text{ for all } i \geq N \text{ for some } N\}, \\ B &= \{x : (x_i) \text{ is bounded}\}, \\ C &= \{x : (x_i) \text{ is unbounded}\} = \mathbb{R}^\omega \setminus B, \\ D &= \{x : (x_i) \text{ converges in } \mathbb{R}\}. \end{aligned}$$

Determine which are open or closed in the product, uniform, and box topologies. Find the product closure of C.

Solution. Every nonempty product-basic open set restricts only finitely many coordinates. Fill the unrestricted tail with zeros to obtain a point of A, hence also of B and D. Fill it instead with the coordinate indices to obtain an unbounded sequence in C. Thus all four sets are dense in the product topology. None is the whole space, so none is closed. None is open either: every product-basic neighborhood meets both B and C; it therefore meets the complements of A and D as well. In particular,

$$\overline{C}^{\,\text{product}} = \mathbb{R}^\omega.$$

In the uniform topology, B and C are both open and closed. Indeed, if $\bar{\rho}(x, y) < 1/2$, then $|x_i - y_i| < 1/2$ for every i, so x is bounded if and only if y is bounded. The same conclusion holds in the finer box topology.

Let

$$c_0 = \{x : x_i \longrightarrow 0\}.$$

The uniform closure of A is c_0. If $x \in c_0$, its truncations $x^{(N)} = (x_1, \ldots, x_N, 0, 0, \ldots)$ satisfy

$$\bar{\rho}(x, x^{(N)}) \leq \sup_{i>N} |x_i| \longrightarrow 0.$$

Conversely, if x lies in the uniform closure of A, then, for any $0 < \varepsilon < 1$, there is $a \in A$ with $\bar{\rho}(x, a) < \varepsilon$. Beyond the finite support of a, $|x_i| < \varepsilon$, so $x_i \to 0$. Hence A is not uniform-closed: $(1/i)$ is in $c_0 \setminus A$. It has empty uniform interior because an arbitrarily small perturbation can make infinitely many tail coordinates nonzero.

The set A is box-closed. If $x \notin A$, then infinitely many x_i are nonzero. At every such coordinate choose an interval about x_i excluding 0. Their product, with arbitrary open neighborhoods at the other coordinates, is a box neighborhood disjoint from A. But A is not box-open: in any basic

neighborhood of an eventually zero sequence, choose a nonzero value at each of infinitely many zero coordinates.

The set D is uniform-closed. Suppose x lies in its uniform closure. Given $\varepsilon > 0$, choose $y \in D$ with

$$\bar{\rho}(x, y) < \min\{1/2, \varepsilon/3\}.$$

Since y converges, there is N such that $|y_i - y_j| < \varepsilon/3$ for $i, j \geq N$. Then

$$|x_i - x_j| \leq |x_i - y_i| + |y_i - y_j| + |y_j - x_j| < \varepsilon.$$

Thus (x_i) is Cauchy and converges in $\mathbb{R}$. It is not uniform-open: if $x_i \to L$, then $y_i = x_i + \eta(-1)^i$, for arbitrarily small $\eta > 0$, does not converge but is uniformly close to x.

Finally, D is box-open and box-closed. Closedness follows from uniform-closedness. If $x_i \to L$, then every point of

$$\prod_{i=1}^{\infty}(x_i - 1/i, x_i + 1/i)$$

also converges to L, proving openness. The results are

Set	Product	Uniform	Box
A	neither	neither	closed, not open
B	neither	open and closed	open and closed
C	neither	open and closed	open and closed
D	neither	closed, not open	open and closed

Chapter 7

Metric Spaces and Uniform Convergence

7.1 Sequences in metric spaces

A sequence converges to x if it is eventually in every neighborhood of x. In a metric space this means that $d(x_n, x) \to 0$.

Proposition 7.1. *Every metric space is Hausdorff.*

Exercise 7.1. Prove that every metric space is Hausdorff.

Solution. For distinct points x, y in a metric space, set $r = d(x, y)/3 > 0$. If z belonged to both $B_d(x, r)$ and $B_d(y, r)$, then $d(x, y) < 2r$, a contradiction. These are disjoint open neighborhoods.

Consequently, the earlier uniqueness result for Hausdorff spaces applies: a sequence in a metric space has at most one limit.

Lemma 7.2 (Sequence lemma)**.** *Let $A \subseteq X$. If a sequence in A converges to x, then $x \in \overline{A}$. If X is metrizable, the converse also holds: every $x \in \overline{A}$ is the limit of a sequence in A.*

Exercise 7.2. Prove the sequence lemma.

Solution. If $a_n \in A$ and $a_n \to x$, every neighborhood of x contains some a_n and therefore meets A. Thus $x \in \overline{A}$.

For the converse, choose a metric inducing the topology. If $x \in \overline{A}$, then $B_d(x, 1/n) \cap A \neq \varnothing$ for every n. Choose a_n in this intersection. Since $d(a_n, x) < 1/n$, the sequence converges to x.

Theorem 7.3. *Let (X, d_X) and (Y, d_Y) be metric spaces. A map $f : X \to Y$ is continuous if and only if, for every $x \in X$ and $\varepsilon > 0$, there is $\delta > 0$ such that*

$$d_X(x, x') < \delta \quad \Longrightarrow \quad d_Y(f(x), f(x')) < \varepsilon.$$

It is also equivalent to require that $x_n \to x$ imply $f(x_n) \to f(x)$ for every sequence in X.

Exercise 7.3. Prove these characterizations of continuity.

Solution. If f is continuous, the inverse image of $B_{d_Y}(f(x), \varepsilon)$ is an open neighborhood of x and contains some $B_{d_X}(x, \delta)$. This gives the stated inequality. Conversely, if the inequality holds and V is open in Y, then for every $x \in f^{-1}(V)$ choose an ε-ball about $f(x)$ contained in V. The corresponding δ-ball about x lies in $f^{-1}(V)$, so this inverse image is open.

Continuity preserves sequence limits in any topological spaces: if V is a neighborhood of $f(x)$, then $f^{-1}(V)$ is a neighborhood of x, and hence eventually contains x_n. For the converse in metric spaces, if continuity fails at x, there is $\varepsilon_0 > 0$ such that for each n one can choose x_n with

$$d_X(x_n, x) < 1/n, \qquad d_Y(f(x_n), f(x)) \geq \varepsilon_0.$$

Then $x_n \to x$ but $f(x_n) \not\to f(x)$, contrary to the assumed sequential condition.

Example 7.4. The box topology on $\mathbb{R}^\omega$ is not metrizable.

Exercise 7.4. Let

$$P = \{x \in \mathbb{R}^\omega : x_i > 0 \text{ for every } i\}.$$

Show that $0 \in \overline{P}$ in the box topology, but no sequence in P converges to 0. Deduce Example 7.4.

Solution. Every box neighborhood of 0 contains a set of the form

$$U = \prod_{i=1}^{\infty} (-\varepsilon_i, \varepsilon_i), \qquad \varepsilon_i > 0.$$

The point $(\varepsilon_i/2)_i$ belongs to $U \cap P$, so $0 \in \overline{P}$.

Suppose $p^{(n)} = (p_i^{(n)})_i$ is any sequence in P. Its diagonal entries $p_n^{(n)}$ are positive. The box neighborhood

$$V = \prod_{i=1}^{\infty} \left(-\frac{p_i^{(i)}}{2}, \frac{p_i^{(i)}}{2} \right)$$

contains none of the terms $p^{(n)}$, because the nth coordinate of $p^{(n)}$ lies outside the nth interval. Consequently $p^{(n)}$ does not converge to 0. This contradicts the sequence lemma if the box topology were induced by a metric.

7.2 Continuous real-valued functions and the diagonal

Lemma 7.5. *If $f, g : X \to \mathbb{R}$ are continuous, then $f + g$, $f - g$, and fg are continuous. If $g(x) \neq 0$ for all x, then f/g is continuous as well.*

Exercise 7.5. Prove the lemma by factoring through product spaces.

Solution. The coordinate criterion shows that $F : X \to \mathbb{R} \times \mathbb{R}$, $F(x) = (f(x), g(x))$, is continuous. Addition and subtraction are continuous, since

$$|(a + b) - (a_0 + b_0)| \leq |a - a_0| + |b - b_0|,$$

and the same estimate holds with a minus sign. Multiplication is continuous because, when $|a - a_0| < 1$,

$$|ab - a_0 b_0| \leq |a|\,|b - b_0| + |b_0|\,|a - a_0| \leq (|a_0| + 1)|b - b_0| + |b_0||a - a_0|.$$

Thus the sums, differences, and products are composites of continuous maps. For $b_0 \neq 0$, the condition $|b - b_0| < |b_0|/2$ gives

$$\left|\frac{1}{b} - \frac{1}{b_0}\right| = \frac{|b - b_0|}{|b|\,|b_0|} \leq \frac{2|b - b_0|}{|b_0|^2}.$$

Therefore reciprocal is continuous on $\mathbb{R} \setminus \{0\}$, and $f/g = f \cdot (1/g)$ is continuous.

The addition construction can be displayed as

$$X \xrightarrow{x \mapsto (f(x), g(x))} \mathbb{R} \times \mathbb{R} \xrightarrow{(a,b) \mapsto a + b} \mathbb{R}$$
$$x \mapsto f(x) + g(x)$$

Exercise 7.6. Suppose $f, g : X \to Y$ are continuous and Y is Hausdorff. Prove that

$$A = \{x \in X : f(x) = g(x)\}$$

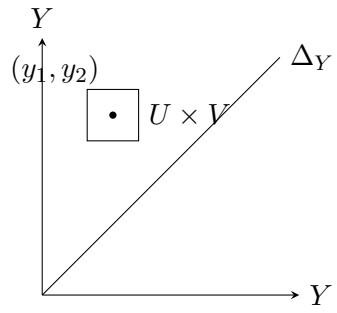
is closed in X.

Solution. Define the diagonal

$$\Delta_Y = \{(y, y) : y \in Y\} \subseteq Y \times Y.$$

If $(y_1, y_2) \notin \Delta_Y$, choose disjoint open neighborhoods U of y_1 and V of y_2. Then $U \times V$ is a neighborhood of (y_1, y_2) disjoint from Δ_Y: a point (y, y) in it would give $y \in U \cap V$. Thus $(Y \times Y) \setminus \Delta_Y$ is open, and Δ_Y is closed.

The map $F : X \to Y \times Y$ defined by $F(x) = (f(x), g(x))$ is continuous, because its coordinate maps are continuous. Since $A = F^{-1}(\Delta_Y)$, the set A is closed.



7.3 Pointwise and uniform convergence

Definition 7.6. Let X be a set and (Y, d) a metric space. A sequence of functions $f_n : X \to Y$ converges *pointwise* to $f : X \to Y$ if

$$\text{for every } x \in X \text{ and } \varepsilon > 0, \text{ there is } N = N(x, \varepsilon) \text{ such that } n \geq N \implies d(f_n(x), f(x)) < \varepsilon.$$

It converges *uniformly* to f if

$$\text{for every } \varepsilon > 0 \text{ there is } N = N(\varepsilon) \text{ such that } n \geq N \implies d(f_n(x), f(x)) < \varepsilon \quad \text{for all } x \in X.$$

The same N must work at every point of X in uniform convergence.

Example 7.7. The continuous functions $f_n : [0,1] \to \mathbb{R}$, $f_n(x) = x^n$, converge pointwise to

$$f(x) = \begin{cases} 0, & 0 \le x < 1, \\ 1, & x = 1. \end{cases}$$

Indeed, for $0 < x < 1$, $n \log x \to -\infty$, so $x^n \to 0$; the endpoints are immediate. The limit is not continuous at 1. Convergence is not uniform: for every n, choose $x \in (0,1)$ with $x^n > 1/2$. Then $|f_n(x) - f(x)| > 1/2$.

For real-valued functions on X, the set of all functions is the product $\mathbb{R}^X$. More generally, for nonempty X and a metric space Y, the formula

$$\bar{\rho}(f,g) = \sup_{x \in X} \min\{d(f(x), g(x)), 1\}$$

defines a uniform metric on Y^X. If X is empty, Y^X has one element and carries its unique metric topology.

Exercise 7.7. Show that $f_n \to f$ in this metric if and only if f_n converges uniformly to f.

Solution. If $\bar{\rho}(f_n, f) \to 0$, choose $0 < \eta < \min\{\varepsilon, 1\}$. For large n,

$$\min\{d(f_n(x), f(x)), 1\} \le \bar{\rho}(f_n, f) < \eta$$

for every x. Since $\eta < 1$, this implies $d(f_n(x), f(x)) < \varepsilon$ uniformly in x.

Conversely, if convergence is uniform, choose $0 < \eta < \varepsilon$. For all sufficiently large n, $d(f_n(x), f(x)) < \eta$ for every x, and therefore

$$\bar{\rho}(f_n, f) \le \eta < \varepsilon.$$

The intermediate choice $\eta < \varepsilon$ accounts for the fact that a supremum of values strictly below η can equal η.

Theorem 7.8 (Uniform limit theorem)**.** *Let X be a topological space and Y a metric space. If continuous functions $f_n : X \to Y$ converge uniformly to $f : X \to Y$, then f is continuous.*

Proof. Let $V \subseteq Y$ be open and $x_0 \in f^{-1}(V)$. Choose $\varepsilon > 0$ with $B_d(f(x_0), \varepsilon) \subseteq V$. Uniform convergence gives N such that

$$d(f_N(x), f(x)) < \varepsilon/3 \qquad \text{for all } x \in X.$$

Continuity of f_N provides an open neighborhood U of x_0 such that

$$f_N(U) \subseteq B_d(f_N(x_0), \varepsilon/3).$$

For $x \in U$, the triangle inequality gives

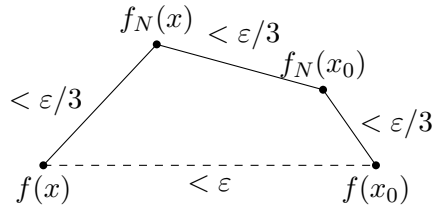
$$\begin{aligned} d(f(x), f(x_0)) &\le d(f(x), f_N(x)) + d(f_N(x), f_N(x_0)) + d(f_N(x_0), f(x_0)) \\ &< \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon. \end{aligned}$$

Thus $f(U) \subseteq V$, so $U \subseteq f^{-1}(V)$. Every point of $f^{-1}(V)$ has such a neighborhood, proving that this inverse image is open.

For real-valued functions the same argument is the decomposition

$$f(x) - f(x_0) = \big(f(x) - f_N(x)\big) + \big(f_N(x) - f_N(x_0)\big) + \big(f_N(x_0) - f(x_0)\big),$$

with each absolute value less than $\varepsilon/3$ near x_0.



Chapter 8

Quotient Spaces

8.1 Quotient maps and quotient topologies

Definition 8.1. Let X and Y be topological spaces. A surjection $p : X \to Y$ is a *quotient map* if, for every $V \subseteq Y$,

$$V \text{ is open in } Y \quad \Longleftrightarrow \quad p^{-1}(V) \text{ is open in } X.$$

In particular, every quotient map is continuous.

Definition 8.2. Let X be a topological space and $p : X \to Y$ a surjection onto a set Y. The *quotient topology* on Y is

$$\mathcal{T}_Y = \{V \subseteq Y : p^{-1}(V) \text{ is open in } X\}.$$

If $\sim$ is an equivalence relation on X, write $X^* = X/\!\sim$ for the set of equivalence classes and give it this topology for the canonical projection

$$p : X \to X^*, \qquad p(x) = [x].$$

Thus $p(x) = p(y)$ exactly when $x \sim y$.

Exercise 8.1. Verify that the quotient topology is a topology. Show that it is the finest topology on Y making p continuous.

Solution. The inverse images of $\varnothing$ and Y are $\varnothing$ and X. Also,

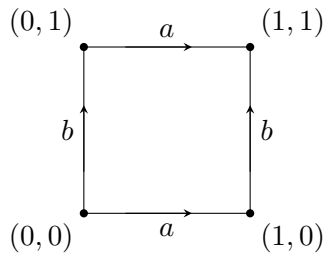
$$p^{-1}\Big(\bigcup_\lambda V_\lambda\Big) = \bigcup_\lambda p^{-1}(V_\lambda), \qquad p^{-1}(V \cap W) = p^{-1}(V) \cap p^{-1}(W).$$

The topology axioms on X give those on Y. By construction p is continuous. If a topology on Y makes p continuous, each of its open sets has open inverse image, and so already belongs to $\mathcal{T}_Y$.

Example 8.3 (The doughnut)**.** Take $X = [0,1] \times [0,1]$. Identify opposite sides by

$$(x,0) \sim (x,1), \qquad (0,y) \sim (1,y) \qquad (0 \le x, y \le 1),$$

and take the equivalence relation generated by these identifications. All four corners lie in one equivalence class. Each other boundary point is identified with the corresponding point on the opposite side, and interior points remain singletons. The resulting quotient is the *torus*, or doughnut.



The arrows specify matching directions along the identified sides.

Note 8.4. A continuous surjection need not be a quotient map. For example, let X be a set with at least two elements, give X the discrete topology, and let Y be the same set with the indiscrete topology. The identity $X \to Y$ is a continuous surjection, but the inverse image of a proper nonempty subset is open even though that subset is not open in Y.

8.2 Open maps and continuous sections

Definition 8.5. A map $f : X \to Y$ is *open* if $f(U)$ is open in Y whenever U is open in X.

Lemma 8.6. *A continuous, open surjection $p : X \to Y$ is a quotient map.*

Exercise 8.2. Prove Lemma 8.6.

Solution. Continuity proves that V open in Y implies $p^{-1}(V)$ open in X. Conversely, if $p^{-1}(V)$ is open, then openness of p and surjectivity give

$$V = p(p^{-1}(V)) \text{ open in } Y.$$

This is exactly the quotient condition.

Example 8.7. For nonempty spaces X and Y, the projection $\pi_X : X \times Y \to X$ is a quotient map. It is a continuous surjection, and it is open: if $W \subseteq X \times Y$ is open and $x \in \pi_X(W)$, choose y with $(x, y) \in W$ and a basic neighborhood $U \times V \subseteq W$. Then $x \in U \subseteq \pi_X(W)$.

Exercise 8.3. Suppose $p : X \to Y$ and $s : Y \to X$ are continuous and

$$p \circ s = \mathrm{id}_Y .$$

Show that p is a quotient map.

Solution. The identity implies $p(s(y)) = y$, so p is surjective. Continuity of p gives one implication of the quotient condition. If $p^{-1}(V)$ is open in X, continuity of s gives

$$s^{-1}(p^{-1}(V)) = (p \circ s)^{-1}(V) = V$$

open in Y, proving the other implication. The map s is a continuous right inverse, also called a continuous section; it need not be an inverse of p in both directions.

Exercise 8.4. Let

$$H = \{(x, y) \in \mathbb{R}^2 : x \geq 0 \text{ or } y = 0\}$$

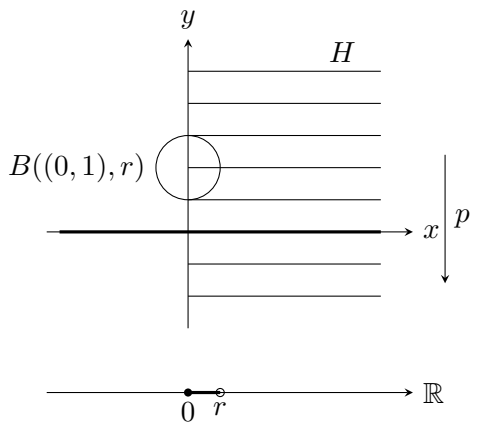
with the subspace topology, and let $p : H \to \mathbb{R}$, $p(x, y) = x$. Show that p is continuous and quotient, but is not open.

Solution. The map is the restriction of a coordinate projection, hence continuous. The map $s : \mathbb{R} \to H$, $s(x) = (x, 0)$, is continuous and satisfies $p \circ s = \mathrm{id}_{\mathbb{R}}$. Exercise 8.3 shows that p is quotient.

Choose $0 < r < 1$ and let $U = H \cap B((0,1), r)$, where B is a Euclidean ball. This is open in H. Its points have positive second coordinate, so they must have first coordinate at least zero. Moreover, $(x, 1) \in U$ for every $0 \leq x < r$. Consequently,

$$p(U) = [0, r),$$

which is not open in $\mathbb{R}$. Thus p is not an open map.



8.3 Maps defined on a quotient

Theorem 8.8. *Let $p : X \to Y$ be a quotient map, and let Z be a topological space. Suppose $g : X \to Z$ is continuous and constant on every fiber $p^{-1}(y)$. Then there is a unique continuous map $\widetilde{g} : Y \to Z$ such that*

$$g = \widetilde{g} \circ p.$$

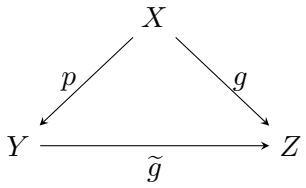
Exercise 8.5. Prove Theorem 8.8.

Solution. For $y \in Y$, choose $x \in p^{-1}(y)$ and define $\widetilde{g}(y) = g(x)$. The fiber condition makes this independent of the choice. The identity $g = \widetilde{g} \circ p$ follows, and surjectivity of p gives uniqueness.

For every open $W \subseteq Z$,

$$p^{-1}(\widetilde{g}^{-1}(W)) = g^{-1}(W)$$

is open in X. The quotient condition makes $\widetilde{g}^{-1}(W)$ open in Y. Hence $\widetilde{g}$ is continuous.



Corollary 8.9. *Under the hypotheses of Theorem 8.8, suppose g is surjective and its fibers are exactly the fibers of p. Then $\widetilde{g}$ is a continuous bijection. It is a homeomorphism if g is a quotient map.*

Exercise 8.6. Prove the corollary.

Solution. Surjectivity of g implies surjectivity of $\widetilde{g}$. If $\widetilde{g}(y) = \widetilde{g}(y')$, choose x, x' with $p(x) = y$ and $p(x') = y'$. Then $g(x) = g(x')$, so equality of the fibers gives $p(x) = p(x')$ and $y = y'$. Thus $\widetilde{g}$ is bijective.

Assume that g is quotient. For an open set $V \subseteq Y$, let $W = \widetilde{g}(V)$. Bijectivity implies

$$g^{-1}(W) = p^{-1}(V),$$

which is open in X. Since g is quotient, W is open in Z. Hence $\widetilde{g}$ is an open continuous bijection, so its inverse is continuous.

Exercise 8.7. Give $\mathbb{R}^2$ the equivalence relation

$$(x_0, y_0) \sim (x_1, y_1) \quad \Longleftrightarrow \quad x_0 + y_0 + y_0^2 = x_1 + y_1 + y_1^2.$$

Show that $\mathbb{R}^2/\!\sim$ is homeomorphic to $\mathbb{R}$. Do the same for the relation defined by $x_0 + y_0^2 = x_1 + y_1^2$.

Solution. Set $g(x, y) = x + y + y^2$. This is continuous and surjective, and its fibers are precisely the equivalence classes. The map

$$s : \mathbb{R} \to \mathbb{R}^2, \qquad s(t) = (t, 0)$$

is continuous and satisfies $g \circ s = \mathrm{id}_{\mathbb{R}}$. Thus g is quotient by Exercise 8.3. If $p : \mathbb{R}^2 \to \mathbb{R}^2/\!\sim$ is the canonical projection, the induced map

$$\widetilde{g} : \mathbb{R}^2/\!\sim \longrightarrow \mathbb{R}, \qquad \widetilde{g}([(x, y)]) = x + y + y^2$$

is a homeomorphism by Corollary 8.9. Its inverse is $t \mapsto [(t, 0)]$. The classes are parabolas $x = t - y - y^2$.

For the second relation use $g_0(x, y) = x + y^2$ and the same section $s(t) = (t, 0)$. The identical argument gives the homeomorphism $[(x, y)] \mapsto x + y^2$, whose fibers are the parabolas $x = t - y^2$.

Chapter 9

Connectedness and the Intermediate Value Theorem

9.1 Separations and examples

Definition 9.1. A *separation* of a topological space X is a pair of disjoint, nonempty open subsets $U, V \subseteq X$ such that $X = U \cup V$. A space is *connected* if it has no separation. A subset of a topological space is connected when it is connected in the subspace topology.

Note 9.2. A partition into two nonempty sets is not enough for a separation: both must be open in the space. For example,

$$\mathbb{R} = (-\infty, a) \cup [a, \infty)$$

is not a separation of $\mathbb{R}$, because $[a, \infty)$ is not open.

Example 9.3. The space $X = [0, 1] \cup [5, 6] \subseteq \mathbb{R}$ is disconnected. Its two intervals are nonempty, disjoint, and open in X; for example,

$$[0, 1] = X \cap (-1, 2), \qquad [5, 6] = X \cap (4, 7).$$

The function

$$f : X \to \mathbb{R}, \qquad f(x) = \begin{cases} 0, & x \in [0, 1], \\ 1, & x \in [5, 6], \end{cases}$$

is continuous, since the inverse image of an open set is one of $\varnothing$, $[0, 1]$, $[5, 6]$, or X. Yet it never takes the value $1/2$. A continuous function on a disconnected domain need not have the intermediate value property.

Example 9.4. The subspace $\mathbb{Q} \subseteq \mathbb{R}$ is disconnected. Since $\sqrt{2} \notin \mathbb{Q}$,

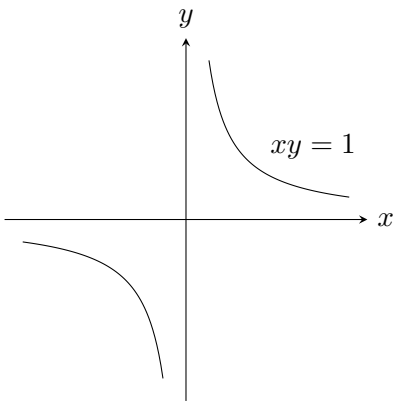
$$U = \mathbb{Q} \cap (-\infty, \sqrt{2}), \qquad V = \mathbb{Q} \cap (\sqrt{2}, \infty)$$

are disjoint nonempty open subsets of $\mathbb{Q}$ whose union is $\mathbb{Q}$.

Example 9.5. The hyperbola $H = \{(x, y) \in \mathbb{R}^2 : xy = 1\}$ is disconnected. The sets

$$H_+ = H \cap \{(x, y) : x > 0\}, \qquad H_- = H \cap \{(x, y) : x < 0\}$$

are nonempty, disjoint, open in H, and cover H, since no point of H has $x = 0$.



9.2 Intervals and continuous images

Theorem 9.6. *Every interval in $\mathbb{R}$ is connected. In particular, $\mathbb{R}$ is connected. Conversely, a connected subset of $\mathbb{R}$ is an interval: if it contains $a < b$, it contains every c with $a < c < b$.*

Exercise 9.1. Prove Theorem 9.6 using the least upper bound property of $\mathbb{R}$.

Solution. Let $I \subseteq \mathbb{R}$ be an interval. Empty and one-point intervals admit no separation. Suppose $I = U \cup V$ were a separation, and choose $a \in U$, $b \in V$, interchanging the two sets if necessary so that $a < b$. The interval property gives $[a, b] \subseteq I$. Let

$$S = U \cap [a, b], \qquad c = \sup S.$$

The set S is nonempty and bounded above, so c exists in $[a, b] \subseteq I$. Since U is relatively open and $a < b$, some points just to the right of a belong to S; thus $c > a$.

If $c \in U$, then $c \neq b$ because $b \in V$. Relative openness of U gives a point $t \in U \cap [a, b]$ with $c < t$, contrary to the upper bound property. If $c \in V$, relative openness of V gives $\delta > 0$ with $(c - \delta, c + \delta) \cap I \subseteq V$. The supremum property gives $s \in S$ with $c - \delta < s \leq c$, contradicting $S \subseteq U$ and $U \cap V = \varnothing$. Both possibilities are impossible, so I is connected.

Conversely, suppose $E \subseteq \mathbb{R}$ contains $a < b$ but omits some $c \in (a, b)$. Then

$$E \cap (-\infty, c), \qquad E \cap (c, \infty)$$

form a separation of E, since they contain a and b, respectively. Thus a connected subset cannot omit such a c.

Proposition 9.7. *The continuous image of a connected space is connected.*

Exercise 9.2. Prove Proposition 9.7.

Solution. Let $f : X \to Y$ be continuous with X connected. Regard f as a continuous surjection $X \to f(X)$, with the subspace topology on its image. If $f(X) = U \cup V$ were a separation, then $f^{-1}(U)$ and $f^{-1}(V)$ would be disjoint nonempty open subsets of X covering X. This contradicts connectedness.

Exercise 9.3. Consider the piecewise bijection $f : [0, 2] \to Y = [0, 1] \cup (2, 3]$ in Example 5.10. Show that its domain is connected and its codomain is disconnected. Use this to give another reason why f cannot be continuous.

Solution. The interval $[0,2]$ is connected. In Y, the sets

$$U = Y \cap (-\infty, 3/2) = [0,1], \qquad V = Y \cap (3/2, \infty) = (2,3]$$

are relatively open, nonempty, and disjoint, and their union is Y. They form a separation, so Y is disconnected.

If f were continuous, its image would be connected by Proposition 9.7. But $f([0,2]) = Y$ because f is surjective, contradicting the separation just exhibited. Therefore f is not continuous.

Theorem 9.8 (Intermediate value theorem)**.** *Let X be connected and $f : X \to \mathbb{R}$ continuous. If $a, b \in X$ and r lies between $f(a)$ and $f(b)$, then $f(c) = r$ for some $c \in X$.*

Exercise 9.4. Prove the intermediate value theorem.

Solution. By Proposition 9.7, $f(X)$ is connected. By Theorem 9.6, $f(X)$ is an interval, so it contains every value between $f(a)$ and $f(b)$. In particular $r \in f(X)$, which means $f(c) = r$ for some $c \in X$.

9.3 Paths and applications of the intermediate value theorem

Definition 9.9. A *path* in X from a to b is a continuous map $\gamma : [0,1] \to X$ with $\gamma(0) = a$ and $\gamma(1) = b$. A space is *path connected* if every pair of its points can be joined by a path.

Exercise 9.5. Show that a path connected space is connected. Show that the circle

$$S^1 = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$$

is path connected.

Solution. Suppose a path connected space X had a separation U, V. Choose $a \in U$ and $b \in V$ and a path γ from a to b. The inverse images $\gamma^{-1}(U)$ and $\gamma^{-1}(V)$ would separate $[0,1]$, contradicting its connectedness.

For S^1, write any two points as $a = (\cos\theta, \sin\theta)$ and $b = (\cos\phi, \sin\phi)$. The path

$$\gamma(t) = \big(\cos((1-t)\theta + t\phi), \sin((1-t)\theta + t\phi)\big)$$

lies in S^1 and joins them. Hence S^1 is path connected, and therefore connected.

Exercise 9.6. Let $f : S^1 \to \mathbb{R}$ be continuous. Show that there exists $p \in S^1$ with $f(-p) = f(p)$.

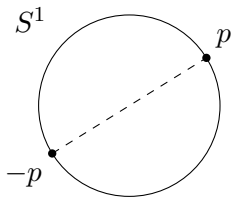
Solution. Define $F : S^1 \to \mathbb{R}$ by

$$F(p) = f(-p) - f(p).$$

The antipodal map $p \mapsto -p$ is continuous, so F is continuous. Also,

$$F(-p) = f(p) - f(-p) = -F(p).$$

Choose $p_0 \in S^1$. If $F(p_0) = 0$, the conclusion follows. Otherwise the values at p_0 and $-p_0$ have opposite signs. Since S^1 is connected, the intermediate value theorem gives a point p with $F(p) = 0$. Equivalently, apply the usual interval theorem to $F \circ \gamma$ on an arc joining p_0 to $-p_0$.



Exercise 9.7 (Fixed point theorem for an interval)**.** Let $f : [0,1] \to [0,1]$ be continuous. Prove that $f(x) = x$ for some $x \in [0,1]$.

Solution. Set $F(x) = f(x) - x$. This function is continuous and satisfies

$$F(0) = f(0) \geq 0, \qquad F(1) = f(1) - 1 \leq 0.$$

If either endpoint value is zero, the corresponding endpoint is a fixed point. Otherwise the values have opposite signs, so the intermediate value theorem gives $x \in (0,1)$ with $F(x) = 0$. Hence $f(x) = x$.

Chapter 10

Compactness and Products

10.1 Closed subspaces and closed intervals

Definition 10.1. An *open cover* of a space X is a family of open subsets of X whose union is X. The space is *compact* if every open cover has a finite subcover. A subset is compact when it is compact with its subspace topology.

Theorem 10.2. *A closed subspace of a compact space is compact.*

Exercise 10.1. Let A be closed in a compact space X. Prove that A is compact.

Solution. Let $\{V_\alpha\}_{\alpha\in I}$ be an open cover of A by sets open in A. Write $V_\alpha = A \cap U_\alpha$, where U_α is open in X. The family

$$\{U_\alpha : \alpha \in I\} \cup \{X \setminus A\}$$

is an open cover of X. Choose a finite subcover. Discarding $X \setminus A$ and intersecting the other members with A gives a finite subcover of A.

Theorem 10.3. *A compact subspace of a Hausdorff space is closed.*

Exercise 10.2. Prove the theorem by separating a point outside the compact subspace from each of its points.

Solution. Let $K \subseteq X$ be compact, with X Hausdorff. If $K = \varnothing$, it is closed. Otherwise, fix $y \in X \setminus K$. For each $x \in K$, choose disjoint open neighborhoods U_x of x and V_x of y. Compactness gives

$$K \subseteq U_{x_1} \cup \cdots \cup U_{x_m}.$$

Then $V = V_{x_1} \cap \cdots \cap V_{x_m}$ is an open neighborhood of y disjoint from K. Thus $X \setminus K$ is open.

Theorem 10.4. *If $a, b \in \mathbb{R}$ and $a \leq b$, then $[a, b]$ is compact.*

Exercise 10.3. Prove compactness of $[a, b]$ using the least upper bound property of $\mathbb{R}$.

Solution. The case $a = b$ is immediate. Suppose $a < b$, and let $\mathcal{U}$ be an open cover of $[a, b]$. Set

$$S = \{t \in [a, b] : [a, t] \text{ is covered by finitely many members of } \mathcal{U}\}.$$

The set S contains a and is bounded above by b. Let $s = \sup S$. Choose $W \in \mathcal{U}$ containing s and $\varepsilon > 0$ such that

$$(s - \varepsilon, s + \varepsilon) \cap [a, b] \subseteq W.$$

There is $t \in S$ with $t > s - \varepsilon$; if $s = a$, take $t = a$. A finite cover of $[a, t]$, together with W, covers $[a, \min\{b, s + \varepsilon/2\}]$. If $s < b$, this gives an element of S larger than s, a contradiction. Thus $s = b$, and the same argument gives a finite cover of $[a, b]$.

10.2 The tube lemma and finite products

Lemma 10.5 (Tube lemma)**.** *Let Y be compact, let $x_0 \in X$, and let W be open in $X \times Y$. If $\{x_0\} \times Y \subseteq W$, then there is an open neighborhood U of x_0 such that $U \times Y \subseteq W$.*

Exercise 10.4. Prove the tube lemma using basic open rectangles.

Solution. If $Y = \varnothing$, take $U = X$. Otherwise, for each $y \in Y$, choose open sets $U_y \subseteq X$ and $V_y \subseteq Y$ with

$$(x_0, y) \in U_y \times V_y \subseteq W.$$

Choose $y_1, \ldots, y_m$ such that $Y = V_{y_1} \cup \cdots \cup V_{y_m}$. Put $U = U_{y_1} \cap \cdots \cap U_{y_m}$. For $(x, y) \in U \times Y$, choose j with $y \in V_{y_j}$. Then $(x, y) \in U_{y_j} \times V_{y_j} \subseteq W$.

Theorem 10.6. *If X and Y are compact, then $X \times Y$, with the product topology, is compact. Consequently, every finite product of compact spaces is compact.*

Exercise 10.5. Use the tube lemma to prove compactness of $X \times Y$.

Solution. If a factor is empty, the product is empty and hence compact. Let $\mathcal{U}$ be an open cover of a nonempty product. For each $x \in X$, the fiber $\{x\} \times Y$ is homeomorphic to Y, so finitely many members of $\mathcal{U}$ cover it. Let W_x be their union. By the tube lemma, some open neighborhood N_x of x satisfies

$$N_x \times Y \subseteq W_x.$$

Finitely many $N_{x_1}, \ldots, N_{x_r}$ cover X. The finitely many cover members used to form $W_{x_1}, \ldots, W_{x_r}$ therefore cover $X \times Y$. Induction gives the finite-product assertion.

10.3 Heine–Borel and extreme values

Definition 10.7. A subset $A \subseteq \mathbb{R}^n$ is *bounded* if there is $M > 0$ such that

$$A \subseteq [-M, M]^n.$$

Equivalently, for every $x = (x_1, \ldots, x_n) \in A$, one has $|x_i| \leq M$ for $1 \leq i \leq n$.

Theorem 10.8 (Heine–Borel theorem)**.** *A subspace A of $\mathbb{R}^n$, with its usual topology, is compact if and only if it is closed in $\mathbb{R}^n$ and bounded.*

Exercise 10.6. Prove both directions of the Heine–Borel theorem using compact intervals and finite products.

Solution. Suppose first that A is compact. Since $\mathbb{R}^n$ is Hausdorff, A is closed in $\mathbb{R}^n$. The open cubes

$$(-m, m)^n, \qquad m \in \mathbb{N},$$

cover A. A finite subcover exists, and the cubes are increasing. Thus $A \subseteq (-M, M)^n \subseteq [-M, M]^n$ for some positive integer M.

Conversely, suppose A is closed and bounded. Choose $M > 0$ with $A \subseteq [-M, M]^n$. Each interval $[-M, M]$ is compact, so the cube $[-M, M]^n$ is compact by the finite-product theorem. The set A is closed in this cube and is therefore compact.

Corollary 10.9 (Extreme-value theorem)**.** *Let X be a nonempty compact space, and let $f : X \to \mathbb{R}$ be continuous. There exist $c, d \in X$ such that*

$$f(c) \leq f(x) \leq f(d) \qquad \text{for all } x \in X.$$

In particular, this holds when $X = [a, b]$ with $a \leq b$.

Exercise 10.7. Show that a continuous image of a compact space is compact, and use this to prove the extreme-value theorem.

Solution. If $\{V_\alpha\}$ is an open cover of $f(X)$ by relatively open sets, write $V_\alpha = f(X) \cap O_\alpha$ with O_α open in the codomain. The sets $f^{-1}(O_\alpha)$ form an open cover of X. A finite subcover gives a finite subcover of $f(X)$.

Thus $K = f(X)$ is a nonempty compact subset of $\mathbb{R}$, so it is closed and bounded. Let $m = \inf K$ and $M = \sup K$. Every neighborhood of m meets K: for every $\varepsilon > 0$, the definition of infimum gives some $t \in K$ with $m \leq t < m + \varepsilon$. Hence $m \in \overline{K} = K$. Similarly $M \in K$. Choose $c, d \in X$ with $f(c) = m$ and $f(d) = M$.

10.4 Arbitrary products

Theorem 10.10 (Tychonoff theorem)**.** *A product $\prod_{\alpha \in I} X_\alpha$ of compact topological spaces is compact in the product topology.*

Exercise 10.8. Prove Tychonoff's theorem using the finite intersection property and Zorn's lemma.

Solution. We use Zorn's lemma in the following form: a nonempty partially ordered set in which every chain has an upper bound has a maximal element. Here the order will be inclusion of families of subsets.

A family has the *finite intersection property* if every finite subfamily has nonempty intersection. By taking complements in the open-cover definition, a space is compact if and only if every family of closed sets with the finite intersection property has nonempty total intersection.

If a factor is empty, the product is empty and compact. The product over the empty index set is a one-point space. Suppose all factors are nonempty, and use the axiom of choice to form the nonempty product $X = \prod_{\alpha \in I} X_\alpha$. Let $\mathcal{F}_0$ be a family of closed subsets of X with the finite intersection property. Zorn's lemma extends $\mathcal{F}_0$ to a maximal family $\mathcal{F}$ of subsets of X with this property: the union of a chain of such families still has the finite intersection property, since any finite selection lies in one member of the chain.

Every finite intersection of members of $\mathcal{F}$ belongs to $\mathcal{F}$. Indeed, adjoining that intersection preserves the finite intersection property, so maximality applies. Also, if a set $E \subseteq X$ meets every member of $\mathcal{F}$, then $E \in \mathcal{F}$, because it meets every finite intersection of members of $\mathcal{F}$.

Fix $\alpha \in I$. The closed sets

$$\{\overline{\pi_\alpha(F)} : F \in \mathcal{F}\}$$

have the finite intersection property in X_α. For a finite selection $F_1, \ldots, F_m$, any point of $F_1 \cap \cdots \cap F_m$ projects to a point in all the displayed closures. Compactness of X_α gives a point

$$x_\alpha \in \bigcap_{F \in \mathcal{F}} \overline{\pi_\alpha(F)}.$$

Choose such a point for every α, and put $x = (x_\alpha)_{\alpha \in I}$.

If $V \subseteq X_\alpha$ is an open neighborhood of x_α, then V meets $\pi_\alpha(F)$ for every $F \in \mathcal{F}$. Thus $\pi_\alpha^{-1}(V)$ meets every $F \in \mathcal{F}$ and belongs to $\mathcal{F}$. Any basic product neighborhood of x is a finite intersection of such sets, so it too belongs to $\mathcal{F}$. It consequently meets every $F \in \mathcal{F}$. Therefore

$$x \in \overline{F} \qquad (F \in \mathcal{F}).$$

In particular, $x \in F$ for every closed set $F \in \mathcal{F}_0$. The original closed family has nonempty intersection, proving compactness.

Chapter 11

Countability and Separation Axioms

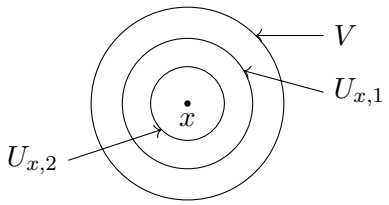
11.1 First and second countability

Definition 11.1. A *countable neighborhood basis at x* is a countable collection of open neighborhoods

$$U_{x,1}, U_{x,2}, \ldots$$

such that every neighborhood V of x contains some $U_{x,k}$. A space is *first countable* if it has a countable neighborhood basis at each point. A space is *second countable* if its topology has a countable basis.

The local basis may be chosen decreasing: replace $U_{x,k}$ by $U_{x,1} \cap \cdots \cap U_{x,k}$. The basis condition is unchanged, since the kth new set is contained in $U_{x,k}$.



Proposition 11.2. *Every second countable space is first countable.*

Exercise 11.1. Obtain a local basis from a countable basis for the topology.

Solution. Let $\mathcal{B}$ be a countable basis. For $x \in X$, the family $\mathcal{B}_x = \{B \in \mathcal{B} : x \in B\}$ is countable. If V is a neighborhood of x, some open set O satisfies $x \in O \subseteq V$. The basis property gives $B \in \mathcal{B}_x$ with $x \in B \subseteq O \subseteq V$.

Theorem 11.3. *Subspaces and countable products of first countable spaces are first countable. Subspaces and countable products of second countable spaces are second countable. Products carry the product topology.*

Exercise 11.2. Prove all four assertions by constructing the required bases.

Solution. Let $Y \subseteq X$. If $\{U_{y,k} : k \in \mathbb{N}\}$ is a local basis at y in X, then $\{Y \cap U_{y,k} : k \in \mathbb{N}\}$ is a local basis at y in Y. To see this, any relatively open neighborhood of y contains a set $Y \cap O$ with O open in X and $y \in O$, and some $U_{y,k}$ is contained in O. Likewise, if $\mathcal{B}$ is a countable basis of X, then

$$\{Y \cap B : B \in \mathcal{B}\}$$

is a countable basis of Y, after empty sets are discarded.

Now let $X = \prod_{n\in\mathbb{N}} X_n$ and fix $x = (x_n) \in X$. Suppose each X_n is first countable, with a local basis $\{U_{n,k} : k \in \mathbb{N}\}$ at x_n. The sets

$$\bigcap_{n\in F} \pi_n^{-1}(U_{n,k_n}), \qquad F \subseteq \mathbb{N} \text{ finite}, \quad k_n \in \mathbb{N},$$

form a local basis at x. There are countably many such sets: there are countably many finite subsets F, and for each F only countably many finite tuples $(k_n)_{n\in F}$. A basic product neighborhood restricts only finitely many coordinates, and a local basis member can be chosen inside each restriction.

If each X_n instead has a countable global basis $\mathcal{B}_n$, use all sets

$$\bigcap_{n\in F} \pi_n^{-1}(B_n), \qquad F \subseteq \mathbb{N} \text{ finite}, \quad B_n \in \mathcal{B}_n.$$

The same counting argument shows that this is countable, and the definition of the product topology shows that it is a basis. Finite products follow by the same construction; an empty product is a one-point space. Empty spaces cause no exception to either assertion.

11.2 The separation axioms

Recall that T_1 means that singletons are closed, and T_2 is the Hausdorff condition.

Definition 11.4. A space is *regular* if it is T_1 and, whenever A is closed and $x \notin A$, there are disjoint open sets containing x and A, respectively. This is the T_3 axiom.

A space is *normal* if it is T_1 and any two disjoint closed sets have disjoint open neighborhoods. This is the T_4 axiom.

Proposition 11.5. *With these conventions,*

$$T_4 \Longrightarrow T_3 \Longrightarrow T_2 \Longrightarrow T_1.$$

Exercise 11.3. Prove the implications between the separation axioms.

Solution. In a normal space, a point and a closed set not containing it are disjoint closed sets, because singletons are closed. Normality therefore gives regularity. In a regular space, if $x \neq y$, separate x from the closed set $\{y\}$ to obtain the Hausdorff property. Finally, the earlier singleton argument shows that every Hausdorff space is T_1.

Theorem 11.6. *Every metrizable space satisfies T_4.*

Proof. Choose a compatible metric d. First, singletons are closed: if $x \neq y$, the ball $B_d(y, d(x, y)/2)$ misses x.

Let A, B be disjoint closed sets. If one is empty, disjoint open neighborhoods are immediate. Otherwise, for every $a \in A$ choose $\varepsilon_a > 0$ with

$$B_d(a, \varepsilon_a) \cap B = \varnothing,$$

and for every $b \in B$ choose $\varepsilon_b > 0$ with

$$B_d(b, \varepsilon_b) \cap A = \varnothing.$$

These choices are possible because the complements of A and B are open. Put

$$U_A = \bigcup_{a \in A} B_d(a, \varepsilon_a/2), \qquad U_B = \bigcup_{b \in B} B_d(b, \varepsilon_b/2).$$

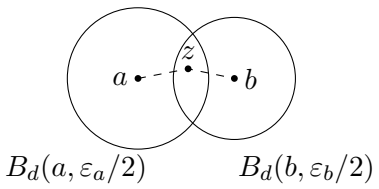
They are open and contain A and B, respectively. Suppose $z \in U_A \cap U_B$. Then for some $a \in A$ and $b \in B$,

$$z \in B_d(a, \varepsilon_a/2) \cap B_d(b, \varepsilon_b/2).$$

If $\varepsilon_b \le \varepsilon_a$, the triangle inequality gives

$$d(a, b) \le d(a, z) + d(z, b) < \frac{\varepsilon_a}{2} + \frac{\varepsilon_b}{2} \le \varepsilon_a.$$

Thus $b \in B_d(a, \varepsilon_a) \cap B$, a contradiction. If $\varepsilon_a \le \varepsilon_b$, the same calculation gives $a \in B_d(b, \varepsilon_b) \cap A$, also a contradiction. Hence $U_A \cap U_B = \varnothing$.



Hypothetical intersection used in the contradiction argument.

11.3 Shrinking open neighborhoods

Regularity and normality allow us to nest open sets with their closures.

Lemma 11.7. *Let X be a T_1 space.*

(a) X is regular if and only if, for every $x \in X$ and open set U containing x, there is an open neighborhood V of x such that $\overline{V} \subseteq U$.

(b) X is normal if and only if, for every closed set $A \subseteq X$ and open set U containing A, there is an open set V such that $A \subseteq V$ and $\overline{V} \subseteq U$.

Proof. For the forward direction of (a), suppose X is regular. The set $A = X \setminus U$ is closed and does not contain x. Choose disjoint open sets V, W such that $x \in V$ and $A \subseteq W$. Since $V \subseteq X \setminus W$ and $X \setminus W$ is closed,

$$x \in V \subseteq \overline{V} \subseteq X \setminus W \subseteq U.$$

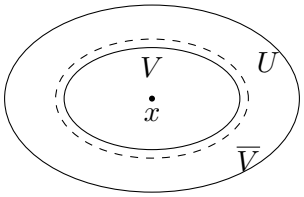
Exercise 11.4. Prove the converse in (a), and prove both directions of (b).

Solution. Suppose the shrinking condition in (a) holds. Given closed A and $x \notin A$, apply it to $U = X \setminus A$. We obtain $x \in V$ and $\overline{V} \subseteq X \setminus A$. The open sets V and $X \setminus \overline{V}$ are disjoint and contain x and A, respectively. Thus X is regular.

For (b), suppose first that X is normal. The closed sets A and $X \setminus U$ are disjoint. Choose disjoint open sets V, W containing them. As before,

$$A \subseteq V \subseteq \overline{V} \subseteq X \setminus W \subseteq U.$$

Conversely, let A, B be disjoint closed sets and apply the shrinking condition to $A \subseteq X \setminus B$. Then $A \subseteq V$ and $\overline{V} \subseteq X \setminus B$. The open sets V and $X \setminus \overline{V}$ separate A and B, so X is normal.



11.4 Regular subspaces and products

Theorem 11.8. *Every subspace of a regular space is regular. Every product of regular spaces is regular in the product topology.*

Exercise 11.5. Prove the theorem using the shrinking characterization of regularity.

Solution. Let $Y \subseteq X$, where X is regular. Singletons are closed in Y because they are intersections of Y with closed singletons in X. Let $y \in Y$ and let O be an open neighborhood of y in Y. Write $O = Y \cap U$, where U is open in X. Choose open $V \subseteq X$ with

$$y \in V \subseteq \overline{V}^{X} \subseteq U.$$

Then $Y \cap V$ is open in Y, contains y, and

$$\overline{Y \cap V}^{Y} \subseteq Y \cap \overline{V}^{X} \subseteq O.$$

The shrinking criterion gives regularity of Y.

Now let $X = \prod_{\alpha \in I} X_\alpha$, with each factor regular. The product is T_1: for any $x \in X$,

$$\{x\} = \bigcap_{\alpha \in I} \pi_\alpha^{-1}(\{x_\alpha\})$$

is closed. Let $x \in O$, where O is open in X. Choose a basic product neighborhood

$$x \in U = \bigcap_{\alpha \in F} \pi_\alpha^{-1}(U_\alpha) \subseteq O, \qquad F \subseteq I \text{ finite.}$$

For every $\alpha \in F$, regularity gives an open set V_α with $x_\alpha \in V_\alpha \subseteq \overline{V_\alpha} \subseteq U_\alpha$. Set $V = \bigcap_{\alpha \in F} \pi_\alpha^{-1}(V_\alpha)$. Each set $\pi_\alpha^{-1}(\overline{V_\alpha})$ is closed and contains V, so

$$x \in V \subseteq \overline{V} \subseteq \bigcap_{\alpha \in F} \pi_\alpha^{-1}(\overline{V_\alpha}) \subseteq U \subseteq O.$$

Again the shrinking criterion proves regularity. Empty products and empty spaces satisfy the assertion as well.

11.5 Second countability and normality

Theorem 11.9. *Every second countable regular space is normal.*

Exercise 11.6. Let A, B be disjoint closed subsets of a second countable regular space. Construct countable open covers of A and B whose members have closures avoiding the other set, and use them to separate A and B.

Solution. If either closed set is empty, the conclusion is immediate. Let $\mathcal{B}$ be a countable basis. For each $a \in A$, regularity gives an open W with $a \in W$ and $\overline{W} \subseteq X \setminus B$. Choose a basis member U with $a \in U \subseteq W$. Then $\overline{U} \cap B = \varnothing$. The collection of all basis members whose closures miss B is countable and covers A. Enumerate it as $U_1, U_2, \ldots$. Similarly obtain $V_1, V_2, \ldots$ covering B, with $\overline{V_n} \cap A = \varnothing$. A finite nonempty list may be repeated to make a sequence.

Define

$$U = \bigcup_{n \geq 1} \left(U_n \setminus \bigcup_{i=1}^{n} \overline{V_i} \right), \qquad V = \bigcup_{n \geq 1} \left(V_n \setminus \bigcup_{i=1}^{n} \overline{U_i} \right).$$

Each term is open, so U, V are open. Every point of A belongs to some U_n and to none of the closures $\overline{V_i}$, hence $A \subseteq U$. Likewise $B \subseteq V$.

Suppose $z \in U \cap V$. There are indices n, m such that

$$z \in U_n \setminus \bigcup_{i \leq n} \overline{V_i}, \qquad z \in V_m \setminus \bigcup_{i \leq m} \overline{U_i}.$$

If $m \leq n$, the first membership excludes $\overline{V_m}$ while the second places z in V_m. If $n \leq m$, the second membership excludes $\overline{U_n}$ while the first places z in U_n. Both are contradictions. Thus $U \cap V = \varnothing$, proving normality.

Chapter 12

Urysohn's Lemma and Metrization

12.1 Separating closed sets by a continuous function

Theorem 12.1 (Urysohn lemma)**.** *Let X be normal, and let $A, B \subseteq X$ be disjoint closed sets. For real numbers $a < b$, there is a continuous function $f : X \to [a, b]$ such that*

$$f(x) = a \quad (x \in A), \qquad f(x) = b \quad (x \in B).$$

Proof. First construct a function $g : X \to [0, 1]$. Let $P = \mathbb{Q} \cap [0, 1]$. Enumerate P without repetitions, beginning with $1, 0$; for instance, continue with

$$\frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{3}{4}, \dots,$$

listing fractions in increasing denominator and omitting those already listed. At each finite stage, order the listed numbers by their usual numerical order.

We construct open sets U_p, for $p \in P$, satisfying

$$p < q \quad \Longrightarrow \quad \overline{U_p} \subseteq U_q. \tag{12.1}$$

Put $U_1 = X \setminus B$. By normality and the shrinking lemma, choose U_0 open with

$$A \subseteq U_0 \subseteq \overline{U_0} \subseteq U_1.$$

Suppose the sets are defined for a finite set P_n containing $0, 1$, and that (12.1) holds for every pair of indices in P_n. Let r be the next rational in the enumeration. In the finite ordered set $P_n \cup \{r\}$, it has an immediate predecessor p and immediate successor q:

$$p < r < q.$$

The sets U_p, U_q are already defined, and $\overline{U_p} \subseteq U_q$. Apply the normal shrinking property to the closed set $\overline{U_p}$ inside the open set U_q. Obtain an open set U_r such that

$$\overline{U_p} \subseteq U_r \subseteq \overline{U_r} \subseteq U_q.$$

The nesting condition holds for all old pairs by induction. If $s \in P_n$ and $s < r$, then $s \leq p$, so

$$\overline{U_s} \subseteq \overline{U_p} \subseteq U_r.$$

If $r < s$, then $s \geq q$, so

$$\overline{U_r} \subseteq U_q \subseteq U_s.$$

Thus the condition holds for every pair in $P_n \cup \{r\}$. Induction defines all the U_p for $p \in P$.

Extend the indexing to all rational numbers by setting

$$U_p = \varnothing \quad (p \in \mathbb{Q},\ p < 0), \qquad U_p = X \quad (p \in \mathbb{Q},\ p > 1).$$

The implication (12.1) still holds for any rational $p < q$.

For $x \in X$, define

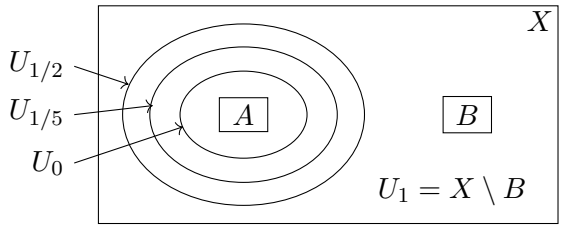
$$Q(x) = \{p \in \mathbb{Q} : x \in U_p\}, \qquad g(x) = \inf Q(x).$$

No negative rational belongs to $Q(x)$, and every rational greater than 1 belongs to $Q(x)$. Thus $Q(x)$ is nonempty and bounded below, and $g(x) \in [0,1]$. If $x \in A$, then $x \in U_0$, so $g(x) = 0$. If $x \in B$, then $x \notin U_1$ and, by nesting, $x \notin U_p$ for any $p \leq 1$. Therefore $Q(x) = \mathbb{Q} \cap (1, \infty)$ and $g(x) = 1$.

The continuity of this infimum function is established in the following exercise. Once it is known, the function

$$f(x) = a + (b - a)g(x)$$

is continuous, takes values in $[a, b]$, and has the required values on A and B.



Exercise 12.1. Suppose U_p is open for every rational p, with $U_p = \varnothing$ for $p < 0$, $U_p = X$ for $p > 1$, and $\overline{U_p} \subseteq U_q$ whenever $p < q$. For

$$g(x) = \inf\{p \in \mathbb{Q} : x \in U_p\},$$

prove that $g : X \to [0,1]$ is continuous.

Solution. For every real t, we claim that

$$\{x : g(x) < t\} = \bigcup_{\substack{p \in \mathbb{Q} \\ p < t}} U_p, \tag{12.2}$$

$$\{x : g(x) > t\} = \bigcup_{\substack{p \in \mathbb{Q} \\ p > t}} (X \setminus \overline{U_p}). \tag{12.3}$$

For (12.2), if $x \in U_p$ with $p < t$, then $g(x) \leq p < t$. Conversely, if $g(x) < t$, the definition of infimum gives an element p of $Q(x)$ with $p < t$; hence $x \in U_p$.

To prove (12.3), suppose first that $x \notin \overline{U_p}$ for some rational $p > t$. For every rational $q < p$, nesting gives $U_q \subseteq \overline{U_q} \subseteq U_p \subseteq \overline{U_p}$. Thus no rational $q < p$ belongs to $Q(x)$, and $g(x) \geq p > t$.

Conversely, if $g(x) > t$, choose rationals p, q with

$$t < p < q < g(x).$$

If $x \in \overline{U_p}$, then $x \in U_q$, implying $g(x) \leq q$, a contradiction. Therefore $x \notin \overline{U_p}$, as required.

The right sides of (12.2) and (12.3) are open. Preimages of the open rays $(-\infty, t)$ and (t, ∞) are therefore open. Since such rays generate the usual topology of $\mathbb{R}$, g is continuous as a real-valued function and hence as a function into $[0, 1]$ with its subspace topology.

12.2 Urysohn's metrization theorem

Theorem 12.2 (Urysohn metrization theorem)**.** *Every second countable regular space is metrizable.*

Exercise 12.2. Prove the metrization theorem by constructing countably many continuous functions that separate points from complements of neighborhoods.

Solution. Let X be second countable and regular. The empty space and a one-point space are metrizable. Assume X has at least two points, and let $\{B_1, B_2, \ldots\}$ be a countable basis. By Theorem 11.9, X is normal.

For each pair (i, j) satisfying $\overline{B_i} \subseteq B_j$, Urysohn's lemma gives a continuous function $f_{ij} : X \to [0, 1]$ with

$$f_{ij} = 1 \text{ on } \overline{B_i}, \qquad f_{ij} = 0 \text{ on } X \setminus B_j.$$

There are only countably many such pairs. Enumerate these functions as $f_1, f_2, \ldots$, repeating functions if the collection is finite.

We first verify the needed separation property. Let $x \in O$, with O open. Choose a basis member B_j with $x \in B_j \subseteq O$. By regularity, choose an open V with $x \in V$ and $\overline{V} \subseteq B_j$. Choose B_i with $x \in B_i \subseteq V$. Then

$$x \in B_i \subseteq \overline{B_i} \subseteq \overline{V} \subseteq B_j \subseteq O.$$

The corresponding function f_{ij} is 1 at x and 0 on $X \setminus O$. In particular, the family separates distinct points: apply this with $O = X \setminus \{y\}$ when $x \neq y$.

Define

$$d(x, y) = \sum_{n=1}^{\infty} 2^{-n} |f_n(x) - f_n(y)|.$$

The series converges because its terms are bounded above by 2^{-n}. Symmetry and nonnegativity are immediate. If $x \neq y$, some f_n distinguishes them, so $d(x, y) > 0$. The triangle inequality follows by summing

$$|f_n(x) - f_n(z)| \leq |f_n(x) - f_n(y)| + |f_n(y) - f_n(z)|$$

over n and passing to the convergent series. Thus d is a metric.

It remains to compare the metric topology with the original topology. Fix $x \in X$ and $\varepsilon > 0$. Choose N such that $\sum_{n>N} 2^{-n} < \varepsilon/2$. By continuity, there is an original open neighborhood W of x on which

$$|f_n(y) - f_n(x)| < \varepsilon/2 \qquad (1 \leq n \leq N).$$

For $y \in W$,

$$d(x, y) < \frac{\varepsilon}{2} \sum_{n=1}^{N} 2^{-n} + \sum_{n>N} 2^{-n} < \varepsilon.$$

Hence every metric neighborhood of a point contains an original open neighborhood of that point. Applying this at each point of a metric-open set shows that every metric-open set is originally open.

Conversely, let O be originally open and $x \in O$. Choose f_n with $f_n(x) = 1$ and $f_n = 0$ on $X \setminus O$. If $y \notin O$, then

$$d(x, y) \geq 2^{-n} |f_n(x) - f_n(y)| = 2^{-n}.$$

Therefore $B_d(x, 2^{-n}) \subseteq O$, so O is metric-open. The two topologies coincide, and X is metrizable.